

A UNIFIED STUDY OF THEORY AND NUMERICAL TECHNIQUES IN THE STEFAN PROBLEM

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for the Degree of

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in

MATHEMATICS

by

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April 2025

DECLARATION

I hereby declare that, this report entitled “**A UNIFIED STUDY OF THEORY AND NUMERICAL TECHNIQUES IN THE STEFAN PROBLEM**” submitted to **Farook College (Autonomous)** towards the partial requirement of **Bachelor of Science in Mathematics**, is an original work carried out by me under the supervision of **Dr. RABEEB ALI V.P.** and has not formed the basis for the award of any degree or diploma, in this or any other institution or university. I have sincerely tried to uphold academic ethics and honesty. Whenever a piece of external information or statement or result is used then, that has been duly acknowledged and cited.

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CERTIFICATE

This is to certify that the work contained in this project report entitled “**A UNIFIED STUDY OF THEORY AND NUMERICAL TECHNIQUES IN THE STEFAN PROBLEM**” submitted by **ANAS MOHAMMED K.T. - FKAWSMT003** to Farook College (Autonomous), Kozhikode towards the partial requirement of **Bachelor of Science in Mathematics** has been carried out by him under my supervision and that it has not been submitted elsewhere for the award of any degree.

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Notations and Abbreviations

$x = s(t)$	The position of the free boundary.
C	The specific heat capacity.
ρ	The density of the specific material.
K	Thermal conductivity.
α	Thermal diffusivity.
t	Latent heat.
Open set	A set which does not contain any of its boundary points.
$\mathbb{R}^n$	The n -dimensional real Euclidean space, where $R = \mathbb{R}^1$.
Du	Gradient vector: $:= \left(\frac{\partial u}{\partial x_1}, \frac{\partial u}{\partial x_2}, \dots, \frac{\partial u}{\partial x_n} \right)$.
$\text{Div } u$	The divergence of a vector-valued function: $:= \sum_{i=1}^n \frac{\partial F_i}{\partial x_i}$.
∂V	The boundary of V .
$\bar{V}$	The closure is defined as $\partial V \cup V$.
C	A function defined on a set is said to be of class C (or C^0) if it is continuous. Thus, C is the class of all continuous functions.
C^K	A function defined on a set is of class C^K if the first K derivatives exist and are continuous.
Smooth	or C^∞ A function is said to be smooth if it has derivatives of all orders.
Test volume	An open and connected subset of $\mathbb{R}^n$ whose boundary is "smooth" enough, but in this essay we demand only C^1 boundary.
$C_{(1)}^2$	A function u is of class $C_{(1)}^2$ when its time derivative u_t is continuous and its spatial derivatives u_x and u_{xx} are continuous.
L	A linear operator acting on functions.
$\text{erf}(x)$	The error function, which is defined as $\text{erf}(f) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} dt$.
$O(h^k)$	Truncation error - An approximation is of order k with respect to h , which means that the error is proportional to h^k .

$\|x\|$ Norm, a measure of length or size of a vector in a vector space.

$\|x\|_\infty$ Maximum norm, the largest value of the vector x .

$\|x\|_2$ l_2 -norm, is defined as $(\sum_{i=1}^n |x_i|^2)^{\frac{1}{2}}$.

Chapter 1

Introduction

The Stefan Problem is a fundamental mathematical model that describes heat transfer and phase-change process, such as melting and solidification. This phenomenon is ubiquitous in nature and various industries, including Material's Science, Physics, Engineering, and Environmental Science. Understanding the Stefan problem is crucial for predicting and controlling the phase-change process, which is essential for optimizing performance, efficiency, and safety in numerous applications.

The Stefan Problem involves solving partial differential equations related to heat distribution in a medium undergoing a phase-change. The main challenge in this problem is that the boundary between the different phases (e.g. solid and liquid) is not known in advance and must be determined as part of the solution. A typical example is the melting of ice and the surrounding water changes position as the melting progresses. Because of the location of the boundary is not fixed, the Stefan problem is often referred to as a 'Free Boundary Problem'.

To obtain a unique solution to a Stefan problem, two boundary conditions are necessary: One to determine the unknown boundary (such as the phase change interface) and another to specify the conditions at the fixed boundary. Although the Stefan problem is most commonly associated with processes like melting and solidification, similar

problems also arises in other areas, such as fluid flow in porous media or shock wave in gas dynamics.

While a general precise formulation for all Stefan problem is not possible, they typically share a few common features: (1) They involve equations describing heat transfer and distributions, (2) There is a clear interface between two (or more) distinct phases, and (3) The temperature at the phase boundary is known in advance.

1.1 Objective

This project aims to develop and analyze a mathematical model for simulating the melting of a solid material, specifically addressing the Stefan problem, and investigate the effects of various parameters on the phase change process. The project develops a mathematical understanding of the Stefan problem with an unknown free boundary in one dimension. This includes explaining the underlying mathematical model, deriving the analytical solution and accessing the validity of the solution. To provide a more comprehensive view, we will also consider two different types of boundary conditions and perform a numerical approximation with the simulated solutions. Given the use of numerical approximations, an error analysis will also be conducted to evaluate the accuracy of the results.

1.2 Historical Note

The Stefan problem is named after the Austrian physicist Joseph Stefan, who first formulated it in 1889. Stefan's work focused on the mathematical modeling of heat transfer during phase changes, specifically the melting of ice. His original formulation dealt with the interface between a solid and liquid (e.g. ice melting into water) and how the heat required for the phase change affects the position of this boundary over time.

The Stefan problem arises from the need to describe how heat diffusion interacts with phase transitions, where the boundary between different phases is not fixed but changes as the material melts or solidifies. Stefan's contribution was groundbreaking because it introduced the concept of a "moving boundary" or "free boundary" where the boundary position is determined by the solution to the problem rather than being predefined. Overtime the Stefan problem expanded beyond its original context of phase changes in materials like ice and water. It has since been applied to various fields.

The generalization of the Stefan problem to more complex scenarios such as multi-phase systems and higher dimensions has been a significant area of research in applied mathematics. In the early 20th century, the Stefan problem gained prominence as the mathematical framework for problem involving heat conduction with phase changes. Researchers built upon Stefan's original work by developing analytical and numerical methods to solve these problems, leading to a rich field of study in both mathematics and physics.

1.3 Physical Background

The Stefan problem describes the heat transfer and phase change in materials, such as the melting of ice and solidification of a liquid. The key feature is that the boundary between the solid and liquid phase is unknown and changes overtime as the phase transition occurs. This moving boundary must be determined as part of the solution.

Phase Change and Heat Transfer

When a substance undergoes a phase change, such as melting or freezing, it requires or releases latent heat at the phase boundary. During this process, the material's temperature remains relatively constant at the phase transition point, even though heat continues to be added or removed. For example, when ice melts, the temperature of

the ice water interface stays at 0°C while heat flows into the system causing the ice to change into liquid water.

This heat flow is governed by the heat conduction equation, which describes how heat is transferred within the material. In regions where there is no phase change, the temperature distribution follows the standard heat equation. However at the phase boundary itself the heat equation must account for the energy required to change the phase of the material, known as latent heat.

Moving Boundary

The key challenge in the Stefan problem is that the interface (boundary) between solid and liquid phases is a priori unknown. The location at the boundary depends on the amount of heat added or removed from the system and must be determined as part of the solution to the problem. The Stefan problem typically involves two boundary conditions: a fixed boundary, and a moving boundary, which changes as the phase transition progresses. Solving this problem requires determining the location of the moving boundary which depends on the heat supplied and the latent heat required to the phase change. To obtain an equation describing the heat distribution, consider any open and piece-wise smooth region Ω where $\Omega \subset \mathbb{R}^n$, with a boundary $\partial\Omega$. By natural assumption (conservation),

rate of change of total quantity = net flux through the boundary $\partial \Omega$

$$\frac{d}{dt} \int_{\Omega} u \, dV = - \int_{\partial\Omega} \mathbf{F} \cdot \boldsymbol{\nu} \, dS = \int_{\Omega} -\text{div} \mathbf{F} \, dV, \quad (1.1)$$

where u is the density of some quantity (such as heat), $\boldsymbol{\nu}$ being the unit normal pointing inwards and $\mathbf{F}$ being the flux density. Assuming that u and u_t exists and are continuous, we can use Leibniz's rule for differentiation under the integral sign on the first integral

in equation 1.1

$$\int_{\Omega} \frac{\partial u}{\partial t} dV = \int_{\Omega} -\text{div} \mathbf{F} dV. \quad (1.2)$$

Now, we present a classical result:

Theorem 1.3.1. *Let f be any function such that $f \in C(\Omega)$ where, $\Omega \subset \mathbb{R}^n$,
If $\int_V f dV = 0$, for all test volumes $V \subset \Omega$, then,*

$$f \equiv 0 \text{ on } \Omega. \quad (1.3)$$

Proof. Assume $f \not\equiv 0$, then at a point $x_0 \in \Omega$, $f(x_0) = \lambda > 0$ (The proof for other case is similar). But since f is continuous we have for any ϵ , for instance $\epsilon = \frac{\lambda}{2}$, there is a $\delta > 0$ such that if

$$|x - x_0| < \delta, x \in \Omega \Rightarrow |f(x) - f(x_0)| < \epsilon = \frac{\lambda}{2}, \quad (1.4)$$

which implies that,

$$\frac{\lambda}{2} < f(x) < \frac{3\lambda}{2}, \quad (1.5)$$

and by integrating the inequality above 1.5 over the test volume V

$$\int_V f(x) dx > \int_V \frac{\lambda}{2} dx = \frac{\lambda}{2} \times \text{volume}(V) > 0. \quad (1.6)$$

Therefore we can conclude that,

$$\int_V f(x) dx > 0. \quad (1.7)$$

we must have a contradiction and thus is $f \equiv 0$.

By writing equation 1.2 as

$$\int_{\Omega} \frac{\partial u}{\partial t} + \text{div} \mathbf{F} dV = 0, \quad (1.8)$$

we must conclude from theorem 1.3.1 that

$$u_t = -\operatorname{div} \mathbf{F}. \quad (1.9)$$

The flux density $\mathbf{F}$ is often proportional to the negative gradient of u (minus since the flow is from higher heat to lower heat):

$$\mathbf{F} = -\alpha Du \ (\alpha > 0), \quad (1.10)$$

where α is the thermal diffusivity. Assuming that α is constant, we get by equation 1.10 and equation 1.9 the *heat equation* :

$$u_t = \alpha \operatorname{div}(Du) = \alpha \Delta u. \quad (1.11)$$

□

Chapter 2

One Dimensional Stefan Problem

Now we will formulate the basic mathematical model describing phase transitions to acquire a solution for the classical Stefan problem (solidification and melting is classical Stefan problem), the heat equation needs to be solved. A boundary condition on the evolving boundary is needed to get a unique solution. It is called "The Stefan Condition"

2.1 Stefan Condition

The Stefan condition is a boundary condition applied to the free boundary in the Stefan problem. It expresses the conservation of energy at the phase change interface, linking the rate of movement of the boundary to the difference in heat fluxes between the two phases. This Stefan condition provides an additional equation necessary to uniquely determine the solution by relating temperature gradients to the rate at which the interface moves.

Assumption (To derive the Stefan condition): The evolving unknown interface is denoted as $X = S(t)$, (" X " is a position in space, " $S(t)$ " is a free boundary and " t " is time)

- As the transitions occur there will be a small volume change (ignore for favour of simplicity).
- By physical reason the temperature should be continuous at the interface $X = S(t)$ between the phases.

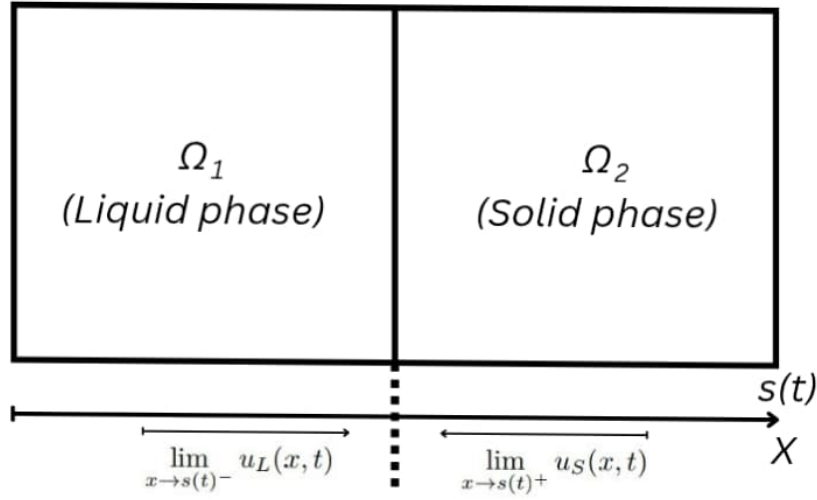


Figure 2.1: (Domain Ω separated into two phases at $X = S(t)$ which are $\Omega_1 = \Omega \cap x < s(t)$ and $\Omega_2 = \Omega \cap x > s(t)$).

$$\lim_{x \rightarrow s(t)^+} u_S(x, t) = \lim_{x \rightarrow s(t)^-} u_L(x, t) = u_m, \quad \text{for all } t. \quad (2.1)$$

The phase-change temperature between the two phases is assumed to be of constant value, u_m . At a fixed time $t = t_0$, consider a domain Ω with two different phases separated at $X = S(t_0)$, which can be seen in Figure (2.1). We assume plane symmetry to have the temperature u to depend on only $stand X$

- When the solid is melting ($t = t_0$) at $X = S(t)$,
 - We should expect that $u \geq u_m$ in liquid phase.
 - $u \leq u_m$ in the solid phase.

Consider a shape of disk $\mathbf{A}$ with area $\mathbf{S}$. Later at time $t_1 > t_0$ the position of the interface has changed to $S(t_1) > S(t_0)$.

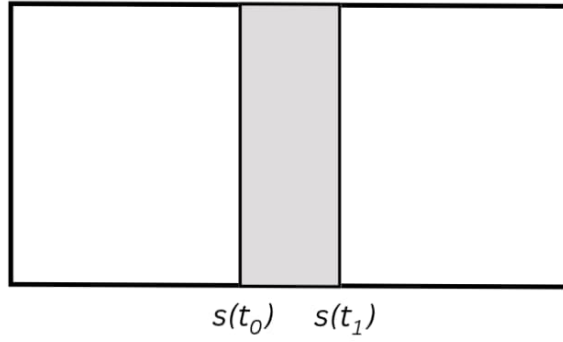


Figure 2.2: The interface initially at $S(t_0)$ shifts to $S(t_1)$ as time increases.

Meanwhile a cylinder of volume $\mathbf{V}$ has melted ;

$$\mathbf{V} = \mathbf{S} \times (S(t_1) - S(t_0)). \quad (2.2)$$

Therefore released a quantity of heat $\mathbf{Q}$;

$$\mathbf{Q} = \mathbf{S} \times (S(t_1) - S(t_0) \times \rho l). \quad (2.3)$$

where l is the specific latent heat and ρ is the density. We known that the heat flux (K_i is material's conductivity) ;

$$\phi_L = -K_L Du_L. \quad (2.4)$$

$$\phi_S = -K_S Du_S. \quad (2.5)$$

We assume $u \in C^1$. By energy conservation it is natural to assume that the total heat absorbed in equation (2.3) is equal to,

$$\begin{aligned} Q &= \int_{t_0}^{t_1} \int_A [\phi_L \cdot \hat{x} + \phi_S \cdot (-\hat{x})] dA d\tau \\ &= \int_{t_0}^{t_1} \int_A [-K_L Du_L(s(\tau), \tau) \cdot \hat{x} - K_S Du_S(s(\tau), \tau) \cdot (-\hat{x})] dA d\tau, \end{aligned} \quad (2.6)$$

Where $\hat{x}$ is the unit vector in the X -direction .

$$\therefore Q = A \int_{t_0}^{t_1} \left[-K_L \frac{\partial u_L}{\partial x}(s(\tau), \tau) + K_S \frac{\partial u_S}{\partial x}(s(\tau), \tau) \right] d\tau. \quad (2.7)$$

We assume this to be equal to expression (2.3). Equating equation (2.3) and (2.7), we get,

$$(s(t_1) - s(t_0))\rho l = \int_{t_0}^{t_1} \left[-K_L \frac{\partial u_L}{\partial x}(s(\tau), \tau) + K_S \frac{\partial u_S}{\partial x}(s(\tau), \tau) \right] d\tau, \quad (2.8)$$

divide by $(t_1 - t_0)$ and take the $\lim_{t_1 \rightarrow t_0}$ ends up with :

$$\rho l \lim_{t_1 \rightarrow t_0} \frac{s(t_1) - s(t_0)}{t_1 - t_0} = \lim_{t_1 \rightarrow t_0} \frac{1}{t_1 - t_0} \int_{t_0}^{t_1} \left[-K_L \frac{\partial u_L}{\partial x}(s(\tau), \tau) + K_S \frac{\partial u_S}{\partial x}(s(\tau), \tau) \right] d\tau. \quad (2.9)$$

Theorem 2.1.1. (*The Intermediate Value Theorem*). Let $f : [a, b] \rightarrow \mathbb{R}$ and $f \in C$, then f attains all values between the endpoints $f(a)$ and $f(b)$.

Theorem 2.1.2. (*The Extreme Value Theorem*). A continuous function f on a bounded and closed interval $[a, b]$, attains both a maximum and a minimum value at least once.

Theorem 2.1.3. (*The Mean-Value Theorem for Integrals*). If $f : [a, b] \rightarrow \mathbb{R}$ and $f \in C$ on $[a, b]$, then there exists a number $c \in [a, b]$ such that,

$$\int_a^b f(x) dx = (b - a)f(c). \quad (2.10)$$

Proof. By Theorem 2.1.2 that a continuous function $f(x)$ has a minimum value m and a maximum value M on an interval $[a, b]$. From the monotonicity of integrals and $m \leq f(x) \leq M$, it follows that,

$$mI = \int_a^b m dx \leq \int_a^b f(x) dx \leq \int_a^b M dx = MI, \quad (2.11)$$

Where,

$$I \equiv \int_a^b dx = b - a, \quad (2.12)$$

equation (2.11) dividing by I (assuming $I > 0$),

$$m \leq \frac{1}{b-a} \int_a^b f(x) dx \leq M. \quad (2.13)$$

From the Extreme Value Theorem we know that both m and M is attained at least once by the integral. Therefore the Intermediate Value Theorem says that the function f attains all values in $[m, M]$, more specific it exists a $c \in [a, b]$ such that,

$$f(c) = \frac{1}{b-a} \int_a^b f(x) dx. \quad (2.14)$$

□

We can with help of Theorem (2.1.3) write equation (2.9) as,

$$l\rho s'(t_1) = \lim_{t_1 \rightarrow t_0} \frac{1}{t_1 - t_0} \times (t_1 - t_0) f(c), \quad (2.15)$$

for simplicity introduce a new function,

$$f(c) \equiv -K_L u_x(s(c), c) + K_S u_x(s(c), c), \quad (2.16)$$

and $c \in [t_0, t_1]$. But as $t_0 \rightarrow t_1$ and f is continuous ($u \in C^1$), then

$$l\rho s'(t_1) = f(t_1). \quad (2.17)$$

However since the same procedure could be done at any time t instead of t_1 , we could instead write,

$$l\rho s'(t) = f(t). \quad (2.18)$$

And hence with our expression for f we reach,

$$l\rho \frac{ds}{dt} = K_S u_x(s(t), t) - K_L u_x(s(t), t). \quad (2.19)$$

Which is called the Stefan condition and is a boundary condition on the free boundary.

2.2 Melting Problem in One Dimension

The Melting Problem in One Dimension deals with phase transitions, specifically the movement of boundaries between phases, such as Solid to Liquid, in response to a heat flow.

- Here we analyze a **semi-infinite solid**, such as a block of ice, exposed to a heat source at one end, causing it to melt.
- This setup is an example of the **Stefan Problem**, a free boundary problem in which the position of the interface between the solid and liquid phase $x = s(t)$, must be determined as part of the solution.
- The system starts with a solid that occupies $x \leq 0$ initially at the melting temperature. Heat is applied at one boundary ($x = 0$), causing the solid to melt and form a liquid phase. This results in two-phase systems.

Liquid Region

The region $0 \leq x \leq s(t)$ contains the melted liquid, with a temperature distribution $u(x, t)$.

Solid Region

The region $s(t) \leq x < \infty$ remains Solid, where the temperature $u(x, t)$ is constant at the melting point.

- We assume the following:
 - The semi-infinite solid occupies the region $0 \leq x < \infty$.
 - The melting occurs at $x = s(t)$ with $s(0) = 0$.
 - The temperature in the liquid phase $u(x, t)$, varies with position and time, while the solid phase remains constant at the melting temperature $u = 0$.
- The goal is to determine:
 - The temperature distribution $u(x, t)$ in the liquid region $0 \leq x < s(t)$.
 - The position of the moving boundary $x = s(t)$, is to be determined.
- To simplify the model:
 - The temperature at the interface $x = s(t)$, remains constant at the melting point.
 - Any volume change during melting is ignored, allowing the focus to remain purely on temperature and boundary movement.
 - Only the temperature distribution in the liquid phase is considered, as the solid is assumed to remain at a fixed temperature equal to or below the melting point.

2.2.1 Heat Equation in the Liquid Phase

As the heat diffuses through the melted layer, the temperature distribution within the liquid phase changes according to the heat equation.

$$\frac{\partial u}{\partial t} = \alpha_L \frac{\partial^2 u}{\partial x^2}, \quad 0 \leq x < s(t), \quad t > 0,$$

$$\alpha_L = \frac{k_L}{C_L \rho},$$

- α_L : Thermal diffusivity of the liquid.
- k_L : Thermal conductivity.
- c_L : Specific heat capacity.
- ρ : Density of the liquid.

This equation shows how the temperature within the liquid evolves over time. The temperature $u(x, t)$ within the liquid region must satisfy the heat equation.

2.2.2 Boundary Condition at $x = 0$

The temperature at the fixed boundary $x = 0$ is given by a heat flux function $f(t)$,

$$u(0, t) = f(t), \quad t > 0, \quad u_0 > 0,$$

where u_0 is the fixed boundary temperature, which is higher than the melting temperature.

Constant Heat Flux

$$f(t) = 1. \tag{2.20}$$

Time-Dependent Heat Flux

$$f(t) = e^t - 1, \quad (2.21)$$

where the boundary temperature increases over time.

Initial Conditions

Initially, we assume that the temperature throughout the solid is below the melting point, so that the entire ice is in a solid state at $x \geq 0$,

$$u(x, 0) = 0, \quad s(0) = 0 \quad \text{initial position of the melting interface.}$$

Stefan Condition at the Interface

At the interface $x = s(t)$ the temperature is constant and fixed at the melting point,

$$u(s(t), t) = 0,$$

we assume that the initial position of the melting front is,

$$s(0) = 0.$$

The motion of the boundary $s(t)$ is governed by Stefan's condition, which ensures energy conservation,

$$l\rho \frac{ds}{dt} = -K_L \frac{\partial u}{\partial x} \bigg|_{x=s(t)}, \quad (2.22)$$

where:

- l is the latent heat of fusion.
- $\frac{ds}{dt}$ is the speed of the moving boundary.
- $\frac{\partial u}{\partial x}$ is the temperature gradient at the interface.
- $-K_L \frac{\partial u}{\partial x}$ is the heat flux in the liquid phase.

The heat carried to the boundary is entirely used to melt the solid.

Solid Phase

$$s(t) \leq x < \infty. \quad (2.23)$$

In the solid region, the temperature remains constant at the melting point,

$$u(x, t) = 0 \quad \forall x \geq s(t), t > 0.$$

This assumption simplifies the problem, as there is no heat conduction in the solid. Therefore, the focus is only on the unknown temperature distribution in the liquid.

2.2.3 Rescaling The Problem into a Dimensionless Form

This section is about transferring a mathematical problem into a dimensionless form. The goal is to simplify the equations by rescaling the variables. So that they become dimensionless, which often makes mathematical analysis easier. Now we define new variables.

- A change of variables is introduced where u and t are rescaled to new variables γ and τ using constraints a and γ .

$$u \rightarrow av \quad (2.24)$$

$$t \rightarrow \gamma\tau \quad (2.25)$$

Where γ and a are constants. The constant a is set to $\gamma^2 = \alpha$ where $\alpha \neq 0$ we can show that the system of equation in (2.23) is The heat equation in the form

$$\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2} \quad (2.26)$$

where $u(x, t)$ represents the temperature at position x and time t . lets start with a general partial differential equation for a function $u(x, t)$,

$$\frac{\partial u}{\partial t} = \alpha \frac{\partial^2 u}{\partial x^2}, \quad (2.27)$$

where α is a constant. To make the equation dimensionless we introduce characteristic scales for time and space. Let

- L be the characteristic length scale.
- T be the characteristic time scale.

We define dimensionless variables for space and time as follows,

$$\zeta = \frac{x}{L}, \tau = \frac{t}{T}. \quad (2.28)$$

Here ζ and τ are dimensionless variables, representing the scaled spatial and temporal coordinates. we express $u(x, t)$ as a function of ζ and τ .

$$u(x, t) = u(\zeta, \tau).$$

where u is now function of dimensionless variables ζ and τ . Now we express the partial derivatives in terms of ζ and τ .

For the time derivative $\frac{\partial u}{\partial t}$:

$$\frac{\partial u}{\partial t} = \frac{\partial u}{\partial \tau} \cdot \frac{d\tau}{dt} = \frac{1}{T} \frac{\partial u}{\partial \tau}. \quad (2.29)$$

For the spatial derivative $\frac{\partial^2 u}{\partial x^2}$,

$$\frac{\partial^2 u}{\partial x^2} = \frac{\partial^2 u}{\partial \zeta^2} \cdot \left(\frac{d\zeta}{dx}\right)^2 = \frac{1}{L^2} \frac{\partial^2 u}{\partial \zeta^2} \quad (2.30)$$

Substitute these equations into the original equation,

$$\frac{1}{T} \frac{\partial u}{\partial \tau} = \alpha \frac{1}{L^2} \frac{\partial^2 u}{\partial \zeta^2}. \quad (2.31)$$

To simplify, we often choose T such that the constants on both sides are balanced.

lets set,

$T = \frac{L^2}{\alpha}$. here T removes α and L from the equation giving,

$$\frac{\partial u}{\partial \tau} = \frac{\partial^2 u}{\partial \zeta^2}, \quad (2.32)$$

Now we have simplified, dimensionless form of the equation becomes,

$$u_\tau = u_{\zeta\zeta}.$$

In this rescaled form, we can drop the capital letters for simplicity, so the equation becomes

$$u_\tau = u_{xx}, \quad (2.33)$$

$$\beta \frac{d}{d\tau} s(\gamma\tau) = -u_x(s(\gamma\tau), \gamma\tau). \quad (2.34)$$

This equation (2.34) express the relationship between the rate of change of $s(\gamma\tau)$ (with respect to τ) and the spatial derivative u_x evaluated at $x = s(\gamma\tau)$ and $t = \gamma\tau$. We know that the initial condition,

$$u(x, 0) = 0 \quad (2.35)$$

$$u(0, \gamma\tau) = f(\gamma\tau)\gamma \quad (2.36)$$

This condition typically derived from a boundary condition in a moving boundary problem (like a Stephan problem), where the rate at which the boundary moves is proportional to a gradient at that boundary. now the parameter $\beta = \frac{lp}{kL}$ is introduced. The functions,

$$s(\gamma, \tau), f(\gamma, \tau), \quad \text{and} \quad v(x, \gamma\tau). \quad (2.37)$$

Are renamed as,

$$\sigma(\tau), f \frac{(T)}{\gamma} \quad \text{and} \quad \lambda(x, \tau). \quad (2.38)$$

Respectively, and now,

$$\lambda\tau(x, \tau) = \lambda(x, \tau), \quad (2.39)$$

if we substitute these rescaled variables and functions into equation (2.34)

we get,

$$\beta \frac{d}{dT} \sigma(\tau) = -\lambda_x(\tau, T), \quad (2.40)$$

however in thesis we make a mathematical approach, so we change the letters,

$$\begin{aligned} \sigma &\rightarrow s \\ \tau &\rightarrow t \\ \lambda &\rightarrow u \\ F &\rightarrow f \end{aligned} \quad (2.41)$$

And we get the rescaled problem as,

$$\begin{aligned} u_t &= u_{xx} \\ u(0, t) &= f(t) \\ u(x, 0) &= 0 \\ \beta \frac{ds}{dt} &= -u_x(s(t), t) \end{aligned} \quad (2.42)$$

Hence, this rescaled form is typically more manageable, as it removes units and makes the problem dimensionless.

2.3 The Similarity Solution

The similarity solution is a method to simplify and solve the heat equation with a moving boundary, which is otherwise challenging due to the dynamic position of the boundary. The similarity solution method transforms the problem from a partial differential equation (PDE) in terms of space and time (x and t) into an ordinary differential equation by a single similarity variable. The set up of the one dimensional

Stefan problem includes the following.

1. The temperature $u(x, t)$ in a semi infinite region $0 \leq x \leq s(t)$ is the position of the phase boundary.
2. The heat equation governing temperature in the liquid phase,

$$\frac{\partial u}{\partial t} = \alpha_L \frac{\partial^2 u}{\partial x^2}, \quad (2.43)$$

where α_L is the thermal diffusivity of the liquid.

Boundary and initial conditions are specified to define the behavior of the temperature at fixed and moving boundaries.

- Boundary at $x = 0$ the temperature is kept constant, so $u(0, t) = u_0 > 0$.
- Boundary at $x = s(t)$ the phase change temperature is reached at this moving boundary, hence $u(s(t), t) = 0$.
- Stefan condition $x = s(t)$: The heat flux at the moving boundary relates to moving the boundary itself.

$$L\rho \frac{\partial S}{\partial t} = -k_L \frac{\partial u}{\partial x} \Big|_{x=S(t)}. \quad (2.44)$$

2.3.1 Similarity Variable Transformation

To simplifying solving this problem, a similarity transformation is applied. The new similarity variable is defined as,

$$\xi = \frac{x}{\sqrt{t}}, \quad (2.45)$$

This change of variable combines spatial and temporal variables into one, there by reducing the problem into an ordinary differential equation. With this transformation

the temperature $u(x, t)$ is assumed to depends only on ξ , not on x and t independently.

$$u(x, t) = F(\xi), \quad (2.46)$$

where $F(\xi)$ is a function of the similarity variable ξ above.

Transforming Heat Equation

By substituting $u(x, t) = F(\xi)$ and using the chain rule we expressed the derivative of u in terms of ξ and the derivative of F .

Time Derivative

$$\frac{\partial u}{\partial t}(x, t) = \frac{\partial F}{\partial t} = \frac{\partial F}{\partial \xi} \cdot \frac{\partial \xi}{\partial t} = -\frac{x}{2t\sqrt{t}} \frac{\partial F}{\partial \xi}. \quad (2.47)$$

Spatial Derivative

$$\frac{\partial u}{\partial x}(x, t) = \frac{\partial F}{\partial \xi} \frac{\partial \xi}{\partial x} = \frac{\partial F}{\partial \xi} \frac{1}{\sqrt{t}}, \quad (2.48)$$

$$\frac{\partial^2 u}{\partial x^2}(x, t) = \frac{1}{t} \frac{\partial^2 F}{\partial \xi^2}, \quad (2.49)$$

substituting gives,

$$-\frac{\xi}{2t} \frac{\partial F}{\partial \xi} = \alpha_L \frac{1}{t} \frac{\partial^2 F}{\partial \xi^2}, \quad (2.50)$$

$$\frac{\partial^2 F}{\partial \xi^2} + \frac{\xi}{2\alpha_L} \frac{\partial F}{\partial \xi} = 0. \quad (2.51)$$

This is an ordinary differential equation that describes the temperature distribution in the liquid. This is a second order linearly ordinary differential equation. To solve it we can try an integrating factor approach.

Rearranging we have,

$$-\frac{\xi}{2\alpha_L} \frac{\partial F}{\partial \xi} = \frac{\partial^2 F}{\partial \xi^2}. \quad (2.52)$$

Let's multiply both sides by an integrating factor $\mu(\xi)$ that will allow at simplify equation,

$$\mu(\xi) \frac{\xi}{2\alpha_L} \frac{\partial F}{\partial \xi} + \mu(\xi) \frac{\partial F}{\partial \xi} = 0. \quad (2.53)$$

To make this a total derivative, choose,

$$\frac{\partial}{\partial \xi}(\mu(\xi)) = \mu(\xi) \frac{\xi}{2\alpha_L}. \quad (2.54)$$

This is separable differential equation, which we can solve by integrating both sides,

$$\int \frac{1}{\mu} d\mu = \int \frac{\xi}{2\alpha_L} d\xi, \quad (2.55)$$

$$\ln \mu = \frac{\xi^2}{4\alpha_L} + c, \quad (2.56)$$

$$\mu(\xi) = e^{\frac{\xi^2}{4\alpha_L}} e^c. \quad (2.57)$$

We can drop e^c as a constant multiplier (since any constant multiple will also work as an integrating factor). So we choose,

$$\frac{\partial}{\partial \xi} \left(e^{\left(\frac{\xi^2}{4\alpha_L}\right)} \frac{\partial F}{\partial \xi} \right) = 0, \quad (2.58)$$

integrating both sides with respect to ξ , we get,

$$e^{\left(\frac{\xi^2}{4\alpha_L}\right)} \frac{\partial F}{\partial \xi} = c_1, \quad (2.59)$$

where c_1 is an integrating constant. Solving for,

$$\frac{\partial F}{\partial \xi} = e^{-\left(\frac{\xi^2}{4\alpha_L}\right)} c_1, \quad (2.60)$$

integrating again to find $F(\xi)$,

$$F(\xi) = c_1 \int e^{-\left(\frac{\xi^2}{4\alpha_L}\right)} d\xi + c_2. \quad (2.61)$$

The integral $\int e^{-\left(\frac{\xi^2}{4\alpha_L}\right)} d\xi$ is related to the error function,

$$\text{erf}(z) = \frac{2}{\sqrt{\pi}} \int_0^z e^{-t^2} dt, \quad (2.62)$$

$$F(\xi) = c_1 \left(\text{erf} \left(\frac{\xi}{2\sqrt{\alpha_L}} \right) \right) + c_2, \quad (2.63)$$

where c_1 and c_2 are constant.

Boundary Condition

1. At $x = 0$ (ξ when $t > 0$) we have $u(0, t) = u_0$. so,

$$F(0) = u_0,$$

$$c_2 = u_0.$$

2. At $x = s(t)$ (or at $\xi = \lambda$ for a constant λ to be determined) we have $u(s(t), t) = 0$ implying,

$$F(\lambda) = 0,$$

$$c_1 = -\frac{\mu_0}{\text{erf}(\xi)},$$

substituting back, we get,

$$F(\lambda) = u_0 - \frac{u_0}{\text{erf}(\xi)} \text{erf} \left(\frac{\xi}{2\sqrt{\alpha_L}} \right), \quad (2.64)$$

which gives the temperature profile in terms of similarity variable ξ , we have,

$$\lambda = \frac{\xi}{\sqrt{\alpha_L}} = \frac{x}{2\sqrt{t\alpha_L}} = \frac{s(t)}{2\sqrt{t\alpha_L}}, \quad (2.65)$$

$$s(t) = 2\lambda\sqrt{\alpha_L t}, \quad (2.66)$$

About The Parameter λ

Stefan condition at the free boundary $x = s(t)$ is,

$$L\rho \frac{\partial s}{\partial t} = -K_L u_x(s(t), \lambda). \quad (2.67)$$

The time derivative of $s(t)$ is,

$$\frac{\partial}{\partial t}(s(t)) = \frac{\partial}{\partial t}(2\lambda\sqrt{\alpha_L t}) = \lambda \frac{\sqrt{\alpha_L}}{\sqrt{t}}, \quad (2.68)$$

$$u_x(x, (t)) = \frac{-u_0}{\operatorname{erf}(\lambda)} \frac{2}{\sqrt{\pi}} \frac{\partial}{\partial x} \left(\int_0^{\frac{x}{2\sqrt{t\alpha_L}}} e^{-u^2} du \right) = \frac{-u_0}{\operatorname{erf}(\lambda)} \frac{e^{\frac{-x^2}{4\alpha_L t}}}{\sqrt{\alpha_L t}}, \quad (2.69)$$

at $x = s(t)$,

$$u_x(s(t), (t)) = \frac{-u_0}{\operatorname{erf}(\lambda)} \frac{e^{-\lambda^2}}{\sqrt{\alpha_L t} \sqrt{\pi}}, \quad (2.70)$$

$$L\rho \frac{\partial s}{\partial t} = -K_L \frac{-u_0}{\operatorname{erf}(\lambda)} \frac{e^{-\lambda^2}}{\sqrt{\alpha_L t} \sqrt{\pi}}, \quad (2.71)$$

$$\frac{\partial s}{\partial t} = \frac{-K_L}{l\rho} \frac{-u_0}{\operatorname{erf}(\lambda)} \frac{e^{-\lambda^2}}{\sqrt{\alpha_L t} \sqrt{\pi}}. \quad (2.72)$$

The phase-change boundary moves according to the similarity solution,

$$s(t) = 2\lambda\sqrt{\alpha_L t}. \quad (2.73)$$

Differentiating with respect to t ,

$$\frac{ds}{dt} = \lambda \frac{\sqrt{\alpha_L}}{\sqrt{t}}. \quad (2.74)$$

Substituting into the Stefan condition,

$$L\rho_L\lambda\frac{\sqrt{\alpha_L}}{\sqrt{t}} = -K_L\frac{\partial u}{\partial x}\Big|_{x=s(t)}. \quad (2.75)$$

From the similarity solution, the temperature gradient at $x = s(t)$ is,

$$\frac{\partial u}{\partial x}\Big|_{x=s(t)} = -\frac{u_0}{\operatorname{erf}(\lambda)}\frac{e^{-\lambda^2}}{\sqrt{\pi\alpha_L t}}. \quad (2.76)$$

Substituting this into the Stefan condition,

$$L\rho_L\lambda\frac{\sqrt{\alpha_L}}{\sqrt{t}} = K_L\frac{u_0}{\operatorname{erf}(\lambda)}\frac{e^{-\lambda^2}}{\sqrt{\pi\alpha_L t}}. \quad (2.77)$$

Rearranging,

$$\lambda e^{\lambda^2} \operatorname{erf}(\lambda) = \frac{K_L}{\rho_L\alpha_L L}u_0\sqrt{\pi}. \quad (2.78)$$

Using the relation $C_L = \frac{K_L}{\rho_L\alpha_L}$,

$$\lambda e^{\lambda^2} \operatorname{erf}(\lambda) = \frac{C_L u_0}{L\sqrt{\pi}}, \quad (2.79)$$

which simplifies to,

$$\lambda e^{\lambda^2} \operatorname{erf}(\lambda) = \frac{st_L}{\sqrt{\pi}}, \quad (2.80)$$

where st_L is the Stefan number, $st_L = \frac{C_L(u_0)}{L}$. st_L is the solution of $u(x, t)$ and $s(t)$, condition for λ , gives

$$\left\{ \begin{array}{l} u(x, t) = u_0 - \frac{u_0}{\operatorname{erf}(\lambda)} \operatorname{erf}\left(\frac{x}{2\sqrt{\alpha_L t}}\right), \\ s(t) = 2\lambda\sqrt{\alpha_L t}, \\ \lambda e^{\lambda^2} \operatorname{erf}(\lambda) = \frac{St_L}{\sqrt{\pi}}. \end{array} \right. \quad (2.81)$$

2.3.2 Non-linearity

If we look at the problem before the free boundary $s(t)$, we have only heat equation. Now, we want to analyse the problem, how it behaves at the free boundary $x = s(t)$, the boundary condition at the free boundary is,

$$-K_L \frac{\partial u}{\partial x}(s(t), t) = L\rho \frac{\partial s}{\partial t}. \quad (2.82)$$

The value at the free boundary is $u(s(t), t) = 0$. For any t where $s(t) \propto \sqrt{t}$ which can be seen in figure,

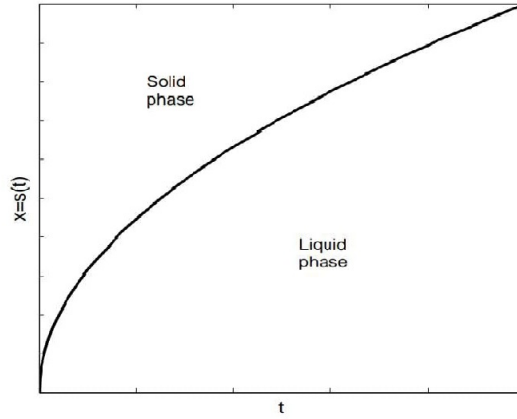


Figure 2.3: The free boundary $s(t)$ against time t .

The curve of free boundary in the (x, t) plane could be described with $r(t) = (x(t), t) = (s(t), t)$.

Velocity vector,

$$v = \frac{\partial r}{\partial t} = \left(\frac{\partial s}{\partial t}, \frac{\partial t}{\partial t} \right) = \left(\frac{\partial s}{\partial t}, 1 \right), \quad (2.83)$$

rate of change of u along the free boundary,

$$v \nabla u = \frac{\partial u}{\partial x} \frac{\partial s}{\partial t} + \frac{\partial u}{\partial t} = 0, \quad (2.84)$$

and that,

$$\frac{\partial u}{\partial x} = - \frac{\frac{\partial u}{\partial t}}{\frac{\partial s}{\partial t}},$$

by multiplying equations we will get.

$$-k_L \left(\frac{\partial u}{\partial x}(s(t), t) \right)^2 = -\rho \frac{\partial u}{\partial t}(s(t), t) = -\frac{LK}{c} \frac{\partial^2 u}{\partial x^2}(s(t), t), \quad (2.85)$$

which implies that the Stefan problem is non linear problem.

2.4 The Maximum Principle

The maximum principle a fundamental tool in analyzing partial differential equations (PDEs), especially those involving heat distribution in phase change problems. The maximum principle states the maximum value of the solution in a domain occurs on the boundary rather than within the interior. This principle is crucial in the Stefan problem, as it allows us to predict temperature behavior without needing exact solutions. The weak and strong forms of the maximum principle are presented, showing that for heat distribution governed by a PDE, interior maximum values imply that the solution is either bound by the boundary or constant across the Domain. This might help establish bounds on the temperature distribution, making it essential for constructing stable and reliable numerical approximations in free boundary problems.

Definition 2.4.1. (The parabolic boundary). The "parabolic boundary" explains a setup commonly used in the study of heat equations. When dealing with heat equations, it is standard to prescribe initial temperature values across a domain and along the domains boundaries.

For a region,

$$E_T = \{(x, t) : 0 < x < l(t), 0 < t \leq T\}, \quad (2.86)$$

in the (x, t) -plane, which represents the space and time dimensions, temperatures are assumed to be known on the boundaries.

$$S_1 : \{x = 0, 0 \leq t \leq T\}, \quad (2.87)$$

$$S_2 : \{0 \leq x \leq l(t), t = 0\},$$

$$S_3 : \{x = l(t), 0 \leq t \leq T\}.$$

Those boundaries together form what is called the parabolic boundary of E_T . This setup helps to establish initial and boundary conditions for solving heat distribution in the defined region.

Lemma 2.4.2. *It states if a function $u(x, t)$ satisfies a certain inequality in a region, if cannot achieve its maximum value within the interior of that region. Specifically if $u(x, t) \in C^2$ (meaning u is twice continuously differentiable) and satisfies the inequality.*

$$L(u) = \frac{\partial^2 u}{\partial x^2} - \frac{\partial u}{\partial t} > 0, \quad (2.88)$$

in the region $E_T = \{(x, t) : 0 < x < l(t), 0 < t \leq T\}$. Then u cannot reach its maximum value at any point within E_T 's interior.

Proof. Assume that u obtain its maximum value at an interior point $P = (x_0, t_0)$ of E_T . As the point P is a critical point, the derivative u_t is 0 and since it is a maximum $u_{xx}(x_0, y_0) \leq 0$. However this contradicts $L[u] > 0$ and thus u cannot have a maximum at an interior point.

□

Theorem 2.4.3. *(Weak Maximum Principle) Suppose a function $u(x, t)$ satisfies a differential inequality of the form,*

$$L[u] = \frac{\partial^2 u}{\partial x^2} - \frac{\partial u}{\partial t} \geq 0, \quad (2.89)$$

in a region $E_T = \{(x, t) : 0 < x < l(t), 0 < t \leq T\}$. Then the maximum value of $u(x, t)$ within the closed region E_T must occur on the boundary of E_T , not in the interior.

Proof. Let M be the maximum value of u on the boundaries of E_T (at $x = 0$, $x = l(t)$ or $t = 0$). Suppose, for contradiction, that u has a higher maximum $M_1 > M$ at some point inside E_T . To test this assumption, we create a function $v(x, t)$, $v(x, t) = u(x, t) + w(x)$, where $w(x)$ is a small adjustment that ensures $v(x, t) < M_1$ on the boundary. This function V now reaches its maximum M_1 at the interior point.

By contradiction, Since $L[u] \geq 0$, we also find that $L[V] > 0$ in E_T . But according to the maximum principle a function satisfying $L[V] > 0$ cannot reach its maximum inside the region. It should occur on the boundary instead. This contradiction means our initial assumption was incorrect. Therefore, u must reach its maximum on the boundary of E_T not inside it. $\square$

Theorem 2.4.4. (*The Strong Maximum Principle*). Suppose that there is a (x_0, t_0) in E_T such that $u(x_0, t_0) = \max_{\bar{E}_T} u$. Then $u(x, t) \equiv u(x_0, t_0)$ for all $(x, t) \in E_T$.

Proof. Assume an interior maximum $u(x, t)$. Suppose $u(x, t)$ reaches its maximum at some interior point $u(x_0, t_0)$ in the domain E_T . At (x_0, t_0) the time derivative of u (specifically u_t) should be zero because the function has stopped increasing. The second spatial derivative u_{xx} is less than or equal to zero at the maximum point because $u(x, t)$ is locally concave down.

- Contradiction with the differential inequality,

For the heat equation, we have $u_t = u_{xx}$. If u reaches its maximum at an interior point, both $u_t = 0$ and $u_{xx} \leq 0$. This contradicts the heat equation because $u_t = u_{xx}$ would imply $u_t \leq 0$ at this point, which prevents an interior maximum from forming. Since an interior maximum contradicts the behaviour prescribed by the heat equation, the maximum value of u cannot occur in the interior unless u is constant throughout the domain. $\square$

Chapter 3

Numerical Method

The study of partial differential equations (PDE's) is essential for numerical analysis. We mainly focus in PDE's is to find solutions that could be expressed explicitly in terms of elementary functions. While there are various methods for finding these solutions to various problems, but some problems lack explicit solution in such cases, numerical methods become valuable for obtaining additional insights into the problem and its solutions.

In the study of differential equations, analysis often contributes by establishing conditions that ensure the uniqueness and existence of solutions themselves, both cases are interesting to study in their own way. When searched for existence and uniqueness in analysis, the problems are posed on infinite dimensional spaces, this contrast to numerical analysis, where problems are discrete and posed on a finite dimensional space. Beyond determining existence and uniqueness for differential discretizations, it is also essential to estimate the error in the approximations.

There is no universal numerical algorithm that can be applied to all PDE's, as different types of numerical schemes are required based on the specific problem being approximated. To approximate the solution of a PDE's is with numerical methods, it is two types,

1. The solution is searched for at a finite number of nodes in the domain of

definition.

2. About expanding the solution as a sum of basis functions defined on the domain of definition and find the coefficients for this expansion.

We will focus exclusively on the first approach, known as the finite difference method.

3.1 Finite Difference Method

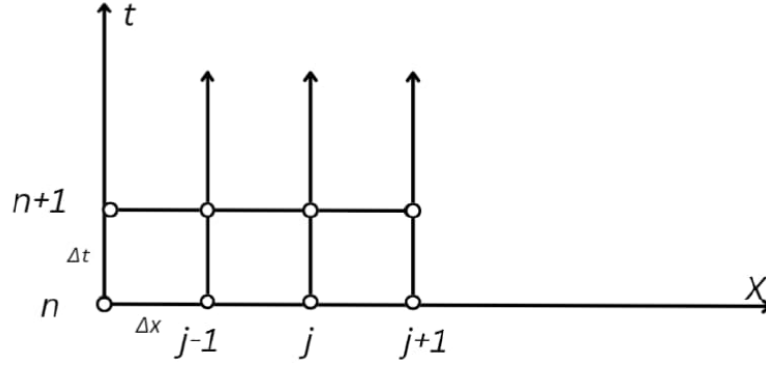
The main idea of the finite difference method is to approximation of the appearing derivatives in a PDE's by sums and difference of function values the function values are defined on the discrete set of points (uniformly spaced nodes), we considering one dimensional heat equation to solve. Two method will be used

- Forward Euler Scheme
- Crank-Nicholson Scheme

3.1.1 Forward Euler Scheme

Forward Euler Scheme is a numerical method for approximating solutions to PDE's this method is explicit since we can express the solution at time step and $n + 1$ in terms of the values of former step n . To demonstrate this method let us start our with a simple example,

$$\begin{aligned}
 u_t &= u_{xx}, & (x, t) &\in (0, 1) \times (0, \infty), \\
 u(x, 0) &= v(x), & \text{in } (0, 1), \\
 u(0, t) &= u(1, t) = 0, & \text{for } t > 0.
 \end{aligned} \tag{3.1}$$



The first step of this procedure is to make space and time discrete. Put,

$$\begin{aligned} x &\rightarrow x_j = x_0 + jh, & j &= 0, 1, 2, \dots, M, \\ t &\rightarrow t^n = t_0 + nk, & n &= 0, 1, 2, \dots \end{aligned} \quad (3.2)$$

Where $x_0 = t_0 = 0$ (it's a initial condition) , M the number of nodes, k the size of the time step and h is spatial step given by,

$$h = \frac{1 - 0}{M} = \frac{1}{M}, \quad (3.3)$$

we denoted the numerical solution U_j^n of u as,

$$U_j^n \approx u(jh, nk). \quad (3.4)$$

Which is the approximative solution of u at time step n and spatial step j . For the spatial derivatives, introduce the forward and backward difference quotients with respect to space and time as,

$$\begin{aligned} \partial_x U_j^n &= \frac{U_{j+1}^n - U_j^n}{h}, & \partial_t U_j^n &= \frac{U_j^{n+1} - U_j^n}{k}. \\ \bar{\partial}_x U_j^n &= \frac{U_j^n - U_{j-1}^n}{h}, & \bar{\partial}_t U_j^n &= \frac{U_j^n - U_j^{n-1}}{k}. \end{aligned} \quad (3.5)$$

∴ Equation (3.1) is written as,

$$\begin{cases} \partial_t U_j^n = \partial_{xx} U_j^n, & \text{for } j = 1, \dots, M-1, n \geq 0, \\ U_0^n = U_M^n = 0, & \text{for } n > 0, \\ U_j^0 = V_j = v(x_j), & \text{for } j = 0, \dots, M. \end{cases} \quad (3.6)$$

Using the figure 3.1.1 solving for U_j^{n+1} in above equation (3.6),

$$\partial_t U_j^n = \partial_x \partial_x U_j^n, \quad (3.7)$$

$$\partial_t U_j^n = \frac{U_j^{n+1} - U_j^n}{k}. \quad (3.8)$$

By Second derivative midpoint formula,

$$\partial_{xx} U_j^n = \frac{U_{j-1}^n - 2U_j^n + U_{j+1}^n}{h^2}, \quad (3.9)$$

form equation (3.6),

$$\begin{aligned} \partial_t U_j^n &= \partial_{xx} U_j^n, \\ \frac{U_j^{n+1} - U_j^n}{k} &= \frac{U_{j-1}^n - 2U_j^n + U_{j+1}^n}{h^2}, \\ U_j^{n+1} &= U_j^n + \frac{k}{h^2} (U_{j-1}^n - 2U_j^n + U_{j+1}^n). \end{aligned} \quad (3.10)$$

$$\lambda = \frac{k}{h^2} \text{ (Mesh ratio),}$$

$$\begin{aligned} U_j^{n+1} &= U_j^n + \lambda (U_{j-1}^n - 2U_j^n + U_{j+1}^n), \\ U_j^{n+1} &= \lambda (U_{j-1}^n + U_{j+1}^n) + (1 - 2\lambda) U_j^n. \end{aligned} \quad (3.11)$$

∴ Equation (3.6) can rewrite into,

$$\begin{aligned} U_j^{n+1} &= \lambda(U_{j-1}^n + U_{j+1}^n) + (1 - 2\lambda)U_j^n, \\ U_0^{n+1} &= U_M^{n+1} = 0, \\ U_j^0 &= V(x_j). \end{aligned} \tag{3.12}$$

The above equation (3.12) in matrix form,

$$\bar{U}^{n+1} = A\bar{U}^n \tag{3.13}$$

Where $\bar{U}^n$ denoted the $(M - 1)$ vector related to U^n and A is a symmetric tridiagonal coefficient matrix given by,

$$A = \begin{bmatrix} 1 - 2\lambda & \lambda & 0 & \cdots & 0 \\ \lambda & 1 - 2\lambda & \lambda & \cdots & 0 \\ 0 & \ddots & \ddots & \ddots & 0 \\ \vdots & \ddots & \lambda & 1 - 2\lambda & \lambda \\ 0 & \cdots & 0 & \lambda & 1 - 2\lambda \end{bmatrix} \tag{3.14}$$

The method is however only first order accurate in time and second order in space.

This implies that the error in the time variable will dominate unless the time spatial step h . The Forward Euler Scheme remains stable who the mesh ratio $\lambda \leq \frac{1}{2}$.

Stability Condition

The stability of the Forward Euler scheme is investigated using the Von Neumann stability method,

$$U_j^n = \xi^n e^{iwjh}. \tag{3.15}$$

Where $i = \sqrt{-1}$, w is a wave number, ξ^n is amplification factor by Forward Euler Scheme, $\lambda = \frac{k}{h^2}$,

$$\xi^{n+1}e^{iwjh} = \xi^n e^{iwjh} + \lambda \xi^n (e^{iw(j-1)h} + 2e^{iwjh} + e^{iw(j+1)h}). \quad (3.16)$$

Divided by $\xi^n e^{iwjh}$,

$$\begin{aligned} G = \xi &= 1 + \lambda(e^{iwh} + 2 + e^{-iwh}), \\ &= 1 + 2\lambda(\cos wh - 1). \end{aligned} \quad (3.17)$$

Put $\cos wh = 1 - 2\sin^2 \frac{wh}{2}$,

$$\begin{aligned} G &= 1 + 2\lambda(1 - 2\sin^2 \frac{wh}{2} - 1), \\ &= 1 - 4\lambda \sin^2 \frac{wh}{2}. \end{aligned} \quad (3.18)$$

The amplification factor $|G| \leq 1$, the numerical solution is stable (by the Von Neumann stability method). At $\sin^2 \frac{wh}{2}$,

$$\begin{aligned} |G| &\leq 1, \\ -1 &\leq G \leq 1, \\ -1 &\leq 1 - 4\lambda \leq 1, \\ -2 &\leq -4\lambda \leq 0, \\ 0 &\leq \lambda \leq \frac{1}{2}. \end{aligned}$$

Thus, the scheme is stable if $0 \leq \lambda \leq \frac{1}{2}$.

Error Analysis

Theorem 3.1.1. *Let U^n and u be the solution of equation (3.1) & (3.6), with $\lambda = \frac{k}{h^2} \leq \frac{1}{2}$ and u being fourth order continuous differentiable, then*

$$\|U^n - u^n\|_{\infty, h} \leq Ct^n h^2 \max_{t \leq t^n} |u(x, t)|_{C^4}, \quad \text{for } t \geq 0,$$

Proof. By Forward Euler scheme approaches the heat equation,

$$u_t = u_{xx},$$

$$\frac{U_j^{n+1} - U_j^n}{k} = \frac{U_{j-1}^n - 2U_j^n + U_{j+1}^n}{h^2}. \quad (3.19)$$

Find the truncation error τ_j^n measure :

$$\tau_j^n = \frac{u(x_j, t_{n+1}) - u(x_j, t_n)}{k} - \frac{u(x_{j+1}, t_n) - 2u(x_j, t_n) + u(x_{j-1}, t_n)}{h^2}. \quad (3.20)$$

By Taylor expansion in time (expand $u(x_j, t_{n+1})$),

$$u(x_j, t_{n+1}) = u(x_j, t_n) + ku_t(x_j, t_n) + \frac{k^2}{2!}u_{tt}(x_j, t_n) + o(k^2), \quad (3.21)$$

$$\therefore \frac{u(x_j, t_{n+1}) - u(x_j, t_n)}{k} = u_t + \frac{k}{2}u_{tt} + o(k^2). \quad (3.22)$$

Taylor expansion in space (expand $u(x_{j+1}, t_n)$ & $u(x_{j-1}, t_n)$),

$$u(x_{j+1}, t_n) = u(x_j, t_n) + hu_x(x_j, t_n) + \frac{h^2}{2}u_{xx} + \frac{h^3}{6}u_{xxx} + o(h^4), \quad (3.23)$$

$$u(x_{j-1}, t_n) = u(x_j, t_n) - hu_x(x_j, t_n) + \frac{h^2}{2}u_{xx} - \frac{h^3}{6}u_{xxx} + o(h^4). \quad (3.24)$$

Substitute $u(x_{j+1}, t_n)$ & $u(x_{j-1}, t_n)$ in,

$$\frac{u(x_{j+1}, t_n) - 2u(x_j, t_n) + u(x_{j-1}, t_n)}{h^2} = u_{xx} + \frac{h^2}{12}u_{xxxx} + o(h^4), \quad (3.25)$$

put equation (3.22) and (3.25) into equation (3.20),

$$\tau = \frac{k}{2}u_{tt} - \frac{h^2}{12}u_{xxxx} + o(k^2) + o(h^4), \quad (3.26)$$

$o(k^2) + o(h^4)$ are negligible when k and h are small. Taking the maximum norm of the τ_j^n ,

$$\|\tau^n\|_\infty, h = \max_j |\tau_j^n| \leq C(k + h^2). \quad (3.27)$$

Because $o(k) + o(h^2)$ is dominating term and C is a depends on the smoothness of $u(x, t)$ particularly $|u(x, t)|_{C^4}$. Now we want to find error,

$$e_j^n = U_j^n - u(x_j, t_n), \quad (3.28)$$

by Forward Euler Scheme at equation (3.29) is leads to,

$$\frac{e_j^{n+1} - e_j^n}{k} = \frac{e_{j-1}^n - 2e_j^n + e_{j+1}^n}{h^2}, \quad (3.29)$$

$$\therefore e_j^{n+1} = e_j^n + \lambda(e_{j-1}^n - 2e_j^n + e_{j+1}^n) + k\tau_j^n. \quad (3.30)$$

The maximum norm measures the largest error across all spatial grid point.

$$\|e^n\|_{\infty, h} = \max_j |e_j^n|, \quad (3.31)$$

form equation (3.30) take the maximum over all j ,

$$\|e^{n+1}\|_{\infty, h} \leq \|e^n\|_{\infty, h} + k\|\tau^n\|_{\infty, h}, \quad (3.32)$$

by iterative error accumulation, the inequality show that the error grows step-by-step,

$$\|e^{n+1}\|_{\infty, h} \leq \|e^n\|_{\infty, h} + \sum_{m=0}^n k\|\tau^m\|_{\infty, h}, \quad (3.33)$$

$$\|e^n\|_{\infty, h} \leq Nk\|\tau^m\|_{\infty, h} \quad \text{where } N = \frac{t_n}{k}, \quad (3.34)$$

$$\|e^n\|_{\infty, h} \leq t_n\|\tau^m\|_{\infty, h}, \quad (3.35)$$

equation (3.1.1) put into equation (3.35),

$$\|e^n\| \leq Ct_n(k + h^2) \quad (3.36)$$

k is small (by $\lambda \leq \frac{1}{2}$ stability condition) *ie*, $o(h^2)$ is dominate $\therefore$ final error bound,

$$\|e^n\|_{\infty, h} \leq Ct_n h^2 \max_{t \leq t_n} |u(x, t)|_{c^4}. \quad (3.37)$$

□

This given us an estimate for the numerical approximation and this numerical scheme we have a quite restrictive stability condition between time step k and spatial step h . With this method it is not possible to have the same order of magnitude in k and h still maintain stability. To give an example of what happens to the numerical simulation with a $\lambda \geq \frac{1}{2}$, we plot the maximum norm against t^n with the Forward Euler Scheme shown in,

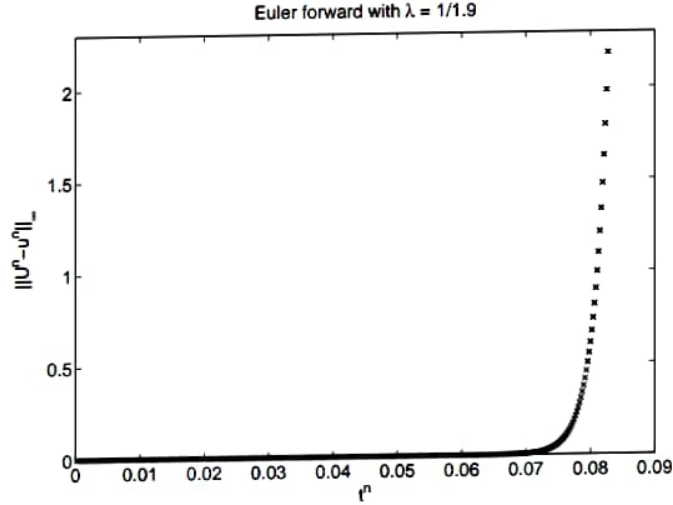


Figure 3.1: This graph plot of the maximum norm of the error against time for the this scheme with $(\lambda > \frac{1}{2})$.

3.1.2 Crank-Nicholson Scheme

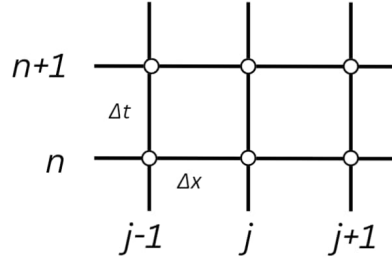
Crank-Nicholson Scheme is a numerical method which is an implicit method, which uses a symmetry around a middle point $(x_j, t_{n=\frac{1}{2}})$, where,

$$t_{n=\frac{1}{2}} = \frac{t^{n+1} + t^n}{k}. \quad (3.38)$$

Due to the implicit scheme an algebraic system must thus be solved for each time step. The example given equation (3.1) with this method is defined as,

$$\begin{cases} \bar{\partial}_t U_j^{n+1} = \frac{1}{2} \partial_x \bar{\partial}_x (U_j^n + U_j^{n+1}), & \text{for } j = 1, 2, \dots, M-1, n \geq 0. \\ U_0^{n+1} = U_M^{n+1} = 0, & \text{for } n > 0. \\ U_j^0 = V_j = v(x_j), & \text{for } j = 0, 1, \dots, M. \end{cases} \quad (3.39)$$

Above equation (3.39) can rewrite,



$$\partial_t U_j^n = \frac{U_j^{n+1} - U_j^n}{k}, \quad (3.40)$$

$$\begin{aligned} \frac{1}{2} \partial_{xx} (U_j^n + U_j^{n+1}) &= c(\partial_x \bar{\partial}_x U_j^n + \partial_x \bar{\partial}_x U_j^{n+1}), \\ &= \frac{1}{2} \left(\frac{U_{j+1}^n - 2U_j^n + U_{j-1}^n}{h^2} + \frac{U_{j+1}^{n+1} - 2U_j^{n+1} + U_{j-1}^{n+1}}{h^2} \right), \end{aligned} \quad (3.41)$$

By condition equation (3.40) equal to (3.41),

$$\frac{U_j^{n+1} - U_j^n}{k} = \frac{1}{2} \left(\frac{U_{j+1}^n - 2U_j^n + U_{j-1}^n + U_{j+1}^{n+1} - 2U_j^{n+1} + U_{j-1}^{n+1}}{h^2} \right), \quad (3.42)$$

$$U_j^{n+1} = U_j^n + \frac{1}{2} \frac{k}{h^2} (U_{j+1}^n - 2U_j^n + U_{j-1}^n + U_{j+1}^{n+1} - 2U_j^{n+1} + U_{j-1}^{n+1}). \quad (3.43)$$

Put $\lambda = \frac{k}{h^2}$,

$$U_j^{n+1} - \frac{\lambda}{2} (U_{j+1}^{n+1} - 2U_j^{n+1} + U_{j-1}^{n+1}) = U_j^n + (U_{j+1}^n - 2U_j^n + U_{j-1}^n), \quad (3.44)$$

$$(1 + \lambda)U_j^{n+1} - \frac{1}{2}\lambda(U_{j+1}^{n+1} + U_{j-1}^{n+1}) = (1 - \lambda)U_j^n + \frac{1}{2}\lambda(U_{j+1}^n + U_{j-1}^n). \quad (3.45)$$

The matrix form is

$$B\bar{U}^{n+1} = A\bar{U}^n. \quad (3.46)$$

Where both A and B are tridaigonal matrices given by :

$$B = \begin{bmatrix} 1 + \lambda & -\frac{1}{2}\lambda & 0 & 0 & \cdots \\ -\frac{1}{2}\lambda & 1 + \lambda & -\frac{1}{2}\lambda & 0 & \cdots \\ 0 & -\frac{1}{2}\lambda & 1 + \lambda & -\frac{1}{2}\lambda & \cdots \\ \vdots & \vdots & \ddots & \ddots & \ddots \\ 0 & \cdots & -\frac{1}{2}\lambda & 1 + \lambda & -\frac{1}{2}\lambda \\ 0 & \cdots & 0 & -\frac{1}{2}\lambda & 1 + \lambda \end{bmatrix},$$

$$A = \begin{bmatrix} 1 - \lambda & \frac{1}{2}\lambda & 0 & 0 & \cdots \\ \frac{1}{2}\lambda & 1 - \lambda & \frac{1}{2}\lambda & 0 & \cdots \\ 0 & \frac{1}{2}\lambda & 1 - \lambda & \frac{1}{2}\lambda & \cdots \\ \vdots & \vdots & \ddots & \ddots & \ddots \\ 0 & \cdots & \frac{1}{2}\lambda & 1 - \lambda & \frac{1}{2}\lambda \\ 0 & \cdots & 0 & \frac{1}{2}\lambda & 1 - \lambda \end{bmatrix}.$$

Solving equation (3.46) we get the solution at time $(n + 1)$,

$$U^{n+1} = B^{-1}AU^n. \quad (3.47)$$

Crank-Nicholson method is second order accurate in time this method give us no time step restriction.

Stability Condition

The stability of the Crank-Nicholson Scheme is checks using the Von-Neumann stability method equation (3.15) :

$$U_j^n = \xi^n e^{iwjh}. \quad (3.48)$$

Where $i = \sqrt{-1}$, w is a wave number, ξ^n is amplification factor by Crank-Nicholson Scheme, $\lambda = \frac{k}{h^2}$,

$$\xi^{n+1} = \xi^n G, \quad (3.49)$$

$$G = \frac{\xi^{n+1}}{\xi^n}, \quad (3.50)$$

$$= \frac{1 - \lambda(1 - \cos(wh))}{1 + \lambda(1 - \cos(wh))}. \quad (3.51)$$

This scheme is unconditionally stable at $|G| \leq 1$ by Von Neumann stability method at $|\cos wh| = 1$,

$$|G| = \frac{1 - \lambda(1 - \cos(wh))}{1 + \lambda(1 - \cos(wh))}, \quad (3.52)$$

$|G| = 1$ is stable always $|G| \leq 1$.

Error Analysis

Theorem 3.1.2. *Let U^n and u be the solution of (3.1) and (3.39) with u being sixth order continuous differentiable, then*

$$\|U^n - u^n\|_{2,h} \leq Ct^n(h^2 + k^2) \max_{t \leq t^n} |u(x, t)|_{C^6}, \text{ for } t \geq 0. \quad (3.53)$$

Proof. For the exact solution $u(x, y)$ substitute Taylor expansion in time and space.

The truncation error is :

$$\tau_j^n = \frac{k^2}{12} u_{ttt}(u_j, t_n) - \frac{h^2}{24} u_{xxxx}(u_j, t_n) + o(k^3) + o(h^4), \quad (3.54)$$

here $o(k^2)$ and $o(h^2)$ is dominated. Taking the L^2 normal of the

$$\|\tau^2\|_{2,h} \leq C(k^2 + h^2) \max_j |u(x, t)|_{C^6}, \quad (3.55)$$

C is a constant and it is depends on the smoothness of $u(x, t)$ particularly $|u(x, t)|_{C^6}$.

Now we want to find error :

$$e_j^n = U_j^n - u(x_j, t_n). \quad (3.56)$$

Then by Crank-Nicholson Scheme given the error propagation equation at $u_t = u_{xx}$,

$$\frac{e_j^{n+1} - e_j^n}{k} = \frac{1}{2} \left(\frac{e_{j+1}^n - 2e_j^n + e_{j-1}^n + e_{j+1}^{n+1} - 2e_j^{n+1} + e_{j-1}^{n+1}}{h^2} \right) + \tau_j^n, \quad (3.57)$$

$$e_j^{n+1} = e_j^n + \frac{1}{2} \frac{k}{h^2} (e_{j+1}^n - 2e_j^n + e_{j-1}^n + e_{j+1}^{n+1} - 2e_j^{n+1} + e_{j-1}^{n+1}) + k\tau_j^n. \quad (3.58)$$

Taking L^2 -norm,

$$\|e^{n+1}\|_{2,h} \leq \|e^n\|_{2,h} + k\|\tau^n\|_{2,h}, \quad (3.59)$$

By iterative this inequality over all the steps n :

$$\|e^{n+1}\|_{2,h} \leq \|e^n\|_{2,h} + \sum_{M=0}^n k\|\tau^M\|_{2,h}, \quad (3.60)$$

$$\|e^n\|_{2,h} \leq Nk\|\tau^n\|_{2,h}. \quad (3.61)$$

Equation (3.55) put into equation (3.61) and $N = \frac{t_n}{k}$,

$$\|e^n\|_{2,h} \leq Ct^n(h^2 + k^2) \max_{t \leq t^n} |u(x, t)|_{C^6}. \quad (3.62)$$

□

Chapter 4

Numerical Analysis

4.1 Analysis When $t \rightarrow 0$

The Stefan Problem is a classic mathematical model for describing phase transitions, such as the melting of ice or the solidification of a liquid. It involves solving the heat equation in a domain whose boundary moves over time. This moving boundary, called the **free boundary**, separates two phases (solid and liquid) and evolves based on the heat transfer at the interface. Here we analyze the behavior of the system as $t \rightarrow 0$. At $t = 0$, the thickness of the melted region, $s(t)$ is zero. This scenario presents a unique challenge as the equations exhibit degeneracy at $t = 0$.

Transforming the Problem

The degeneracy at $t = 0$, where $s(t) \rightarrow 0$, makes the equations difficult to solve. To handle this, we introduce a similarity transformation using dimensionless variables.

Define:

$$\xi = \frac{x}{s(t)}, \quad u = h(t)F(\xi, t) \tag{4.1}$$

where:

- $\xi = \frac{x}{s(t)}$: A scaled spatial variable, normalized to the domain $0 \leq \xi \leq 1$.

- $F(\xi)$: A function describing the temperature profile in ξ .
- $h(t)$: A time-dependent scaling factor as $t \rightarrow 0$.

This transformation simplifies the spatial and time-dependent derivatives in the problem, re-expressing the equation in terms of ξ and F , which can be managed as t becomes small.

- The domain $0 \leq x \leq s(t)$ becomes fixed as $0 \leq \xi \leq 1$.
- The reformulated equations become simpler to analyze.

We express the derivatives of $u(x, t) = h(t)F(\xi, t)$ with respect to x and t as follows.

First Spatial Derivative

$$\frac{\partial u}{\partial x} = h(t) \frac{\partial F}{\partial \xi} \quad (4.2)$$

Since $\xi = \frac{x}{s(t)}$, we have,

$$\frac{\partial \xi}{\partial x} = \frac{1}{s(t)} \quad (4.3)$$

$$\frac{\partial u}{\partial x} = \frac{h(t)}{s(t)} \frac{\partial F}{\partial \xi} \quad (4.4)$$

Second Spatial Derivative

$$\frac{\partial^2 u}{\partial x^2} = \frac{\partial}{\partial x} \left(\frac{h(t)}{s(t)} \frac{\partial F}{\partial \xi} \right) \quad (4.5)$$

Since $h(t)$ and $s(t)$ depend only on t and are independent of x , they can be treated as constants with respect to x .

$$\frac{\partial^2 u}{\partial x^2} = \frac{h(t)}{s(t)} \frac{\partial}{\partial x} \left(\frac{\partial F}{\partial \xi} \right) \frac{1}{s(t)} \quad (4.6)$$

$$= \frac{h(t)}{s(t)^2} \frac{\partial^2 F}{\partial \xi^2} \quad (4.7)$$

$$\frac{\partial^2 u}{\partial x^2} = \frac{h(t)}{s(t)^2} \frac{\partial^2 F}{\partial \xi^2} \quad (4.8)$$

Time Derivative

Let

$$u(x, t) = h(t)F(\xi, t). \quad (4.9)$$

Applying the product rule,

$$\frac{\partial u}{\partial t} = \frac{dh}{dt}F + h(t)\frac{\partial F}{\partial t}. \quad (4.10)$$

Since F depends on t both directly and indirectly (through ξ), we expand $\frac{\partial F}{\partial t}$ using the chain rule,

$$\frac{\partial F}{\partial t} = \frac{\partial F}{\partial t} + \frac{\partial F}{\partial \xi} \frac{\partial \xi}{\partial t}. \quad (4.11)$$

Since

$$\xi = \frac{x}{s(t)}, \quad (4.12)$$

differentiating ξ with respect to t gives,

$$\frac{\partial \xi}{\partial t} = -\frac{x}{s(t)^2} \frac{ds}{dt} = -\frac{\xi}{s(t)} \frac{ds}{dt}. \quad (4.13)$$

Substituting this into our expression for $\frac{\partial u}{\partial t}$:

$$\frac{\partial u}{\partial t} = \frac{dh}{dt}F + h(t) \left(\frac{\partial F}{\partial t} - \frac{\xi}{s(t)} \frac{ds}{dt} \frac{\partial F}{\partial \xi} \right). \quad (4.14)$$

Substituting into the Heat Equation

The heat equation is given by:

$$\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2} \quad (4.15)$$

Substituting $u = h(t)F(\xi)$, the equation becomes:

$$\frac{d}{dt}(hF) = \frac{\partial^2}{\partial x^2}(hF) \quad (4.16)$$

Expanding both sides:

$$\frac{dh}{dt}F + h(t)\frac{\partial F}{\partial t} = \frac{h(t)}{s(t)^2} \frac{\partial^2 F}{\partial \xi^2} \quad (4.17)$$

Equating both sides:

$$\frac{dh}{dt}F + h(t) \left(\frac{\partial F}{\partial t} - \frac{\xi}{s(t)} \frac{ds}{dt} \frac{\partial F}{\partial \xi} \right) = \frac{h(t)}{s(t)^2} \frac{\partial^2 F}{\partial \xi^2} \quad (4.18)$$

Multiplying both sides by s^2 :

$$s(t)^2 \frac{dh}{dt}F + s(t)^2 h(t) \frac{\partial F}{\partial t} - \xi s(t) h(t) \frac{ds}{dt} \frac{\partial F}{\partial \xi} = h(t) \frac{\partial^2 F}{\partial \xi^2} \quad (4.19)$$

Rearranging:

$$h \frac{\partial^2 F}{\partial \xi^2} = s(t) \left[s(t) \frac{dh}{dt}F + s(t) h \frac{\partial F}{\partial t} - \xi \frac{ds}{dt} \frac{\partial F}{\partial \xi} \right] \quad (4.20)$$

4.1.1 Constant Boundary Condition

A constant boundary condition is applied at $x = 0$, meaning the temperature at the boundary is fixed. To ensure F is independent of t as $t \rightarrow 0$, $h(t)$ is chosen as a constant, $h(t) = 1$, which ensures $F(0) = 1$. Substituting $h(t) = 1$ into the heat equation:

$$\frac{\partial^2 F}{\partial \xi^2} = s(t) \left(s(t) \frac{d}{dt}(1 \cdot F) + s(t) \frac{\partial F}{\partial t} - \xi s(t) \frac{ds}{dt} \frac{\partial F}{\partial \xi} \right) \quad (4.21)$$

Since $h(t) = 1$, this simplifies to:

$$\frac{\partial^2 F}{\partial \xi^2} = s(t)^2 \frac{\partial F}{\partial t} - \xi s(t) \frac{ds}{dt} \frac{\partial F}{\partial \xi} \quad (4.22)$$

Let:

$$s(t) = 2\lambda\sqrt{\alpha t}, \quad \text{where } \alpha = 1 \quad (4.23)$$

Thus,

$$s(t) = 2\lambda\sqrt{t} \quad (4.24)$$

Differentiating:

$$\frac{ds}{dt} = \frac{d}{dt}(2\lambda\sqrt{t}) = \frac{\lambda}{\sqrt{t}} \quad (4.25)$$

Substituting $s(t)$ and $\frac{ds}{dt}$ into the equation:

$$\frac{\partial^2 F}{\partial \xi^2} = (2\lambda\sqrt{t})^2 \frac{\partial F}{\partial t} - \xi(2\lambda\sqrt{t}) \frac{\lambda}{\sqrt{t}} \frac{\partial F}{\partial \xi} \quad (4.26)$$

Since:

$$s^2 = (2\lambda\sqrt{t})^2 = 4\lambda^2 t \quad (4.27)$$

$$\xi s \frac{ds}{dt} = \xi(2\lambda\sqrt{t}) \frac{\lambda}{\sqrt{t}} = 2\lambda^2 \xi \quad (4.28)$$

Rewriting:

$$s^2 \frac{\partial F}{\partial t} = 2\lambda^2 \xi \frac{\partial F}{\partial \xi} \quad (4.29)$$

As $t \rightarrow 0$, $F(\xi, t)$ becomes independent of t , meaning:

$$\frac{\partial F}{\partial t} = 0 \quad (4.30)$$

Thus, the equation reduces to,

$$\frac{\partial^2 F}{\partial \xi^2} = -s\xi \frac{ds}{dt} \frac{\partial F}{\partial \xi} \quad (4.31)$$

Substituting values,

$$\frac{\partial^2 F}{\partial \xi^2} = -(2\lambda\sqrt{t})\xi \left(\frac{\lambda}{\sqrt{t}} \right) \frac{\partial F}{\partial \xi} \quad (4.32)$$

$$= -(2\lambda\xi)(\lambda) \frac{\partial F}{\partial \xi} \quad (4.33)$$

$$= -2\lambda^2 \xi \frac{\partial F}{\partial \xi} \quad (4.34)$$

4.1.2 Boundary Conditions for $F(\xi, t)$

We now establish boundary conditions based on the constant boundary temperature conditions:

1. The temperature at the moving interface $x = s(t)$ remains fixed at the melting

temperature:

$$u(s(t), t) = 0, \quad \forall t > 0. \quad (4.35)$$

Since $\xi = \frac{x}{s(t)}$, the moving boundary $x = s(t)$ corresponds to $\xi = 1$, which implies:

$$F(1) = 0. \quad (4.36)$$

2. The fixed boundary at $x = 0$ is maintained at a constant temperature:

$$u(0, t) = f(t) = 1, \quad \forall t \geq 0. \quad (4.37)$$

The temperature at $x = 0$ corresponds to $\xi = 0$, and since

$$u(x, t) = h(t)F(\xi), \quad (4.38)$$

we have,

$$u(0, t) = h(t)F(0). \quad (4.39)$$

Given $f(t) = 1$, it follows that:

$$F(0) = \frac{1}{h(t)}. \quad (4.40)$$

Choosing $h(t) = 1$, we obtain:

$$F(0) = 1. \quad (4.41)$$

3. The Stefan condition is derived from the conservation of energy at the moving boundary. It relates the rate of movement of the interface $s(t)$ to the heat flux at the boundary.

4.1.3 Stefan Condition

The Stefan condition, which expresses energy conservation at the moving boundary $x = s(t)$, is given by:

$$\rho L \frac{ds}{dt} = -k \frac{\partial u}{\partial x} \Big|_{x=s(t)} \quad (4.42)$$

Since we have:

$$\frac{ds}{dt} = \frac{\lambda}{\sqrt{t}} \quad (4.43)$$

and

$$\frac{\partial u}{\partial x} \Big|_{x=s(t)} = \frac{h(t)}{s(t)} \frac{\partial F}{\partial \xi} \Big|_{\xi=1} \quad (4.44)$$

Substituting $s(t) = 2\lambda\sqrt{t}$ and $h(t) = 1$, we get:

$$\rho L \frac{\lambda}{\sqrt{t}} = -k \frac{1}{2\lambda\sqrt{t}} \frac{\partial F}{\partial \xi} \Big|_{\xi=1} \quad (4.45)$$

Simplifying,

$$\rho L \lambda = -k \frac{1}{2\lambda} \frac{\partial F}{\partial \xi} \quad (4.46)$$

Multiplying both sides by 2λ ,

$$2\rho L \lambda^2 = -k \frac{\partial F}{\partial \xi} \quad (4.47)$$

Defining $\beta = \frac{\rho L}{k}$, we obtain:

$$2\lambda^2 \beta = -\frac{\partial F}{\partial \xi} \Big|_{\xi=1} \quad (4.48)$$

Thus, the Stefan condition can be rewritten as:

$$\beta s(t) \frac{ds}{dt} = -h \frac{\partial F}{\partial \xi} \Big|_{\xi=1} \quad (4.49)$$

Since $h(t) = 1$, this simplifies to:

$$\beta s(t) \frac{ds}{dt} = -\frac{\partial F}{\partial \xi} \Big|_{\xi=1} \quad (4.50)$$

Boundary Conditions

$$F(1) = 0, \quad (4.51)$$

$$F(0) = 1, \quad (4.52)$$

$$2\lambda^2\beta = -\frac{\partial F}{\partial \xi} \Big|_{\xi=1}. \quad (4.53)$$

Differential Equation and Boundary Conditions

The governing differential equation is:

$$\frac{\partial^2 F}{\partial \xi^2} = -2\lambda^2 \xi \frac{\partial F}{\partial \xi}, \quad 0 \leq \xi \leq 1. \quad (4.54)$$

Rewriting the equation,

$$\frac{\partial^2 F}{\partial \xi^2} + 2\lambda^2 \xi \frac{\partial F}{\partial \xi} = 0. \quad (4.55)$$

Multiplying through by the integrating factor $e^{\lambda^2 \xi^2}$,

$$e^{\lambda^2 \xi^2} \frac{\partial^2 F}{\partial \xi^2} + 2\lambda^2 \xi e^{\lambda^2 \xi^2} \frac{\partial F}{\partial \xi} = 0. \quad (4.56)$$

Recognizing this as a derivative,

$$\frac{\partial}{\partial \xi} \left(e^{\lambda^2 \xi^2} \frac{\partial F}{\partial \xi} \right) = 0. \quad (4.57)$$

Integrating once,

$$e^{\lambda^2 \xi^2} \frac{\partial F}{\partial \xi} = c_1, \quad (4.58)$$

where c_1 is an integration constant. For the second integration, rearrange for $\frac{\partial}{\partial \xi}$,

$$\frac{\partial F}{\partial \xi} = c_1 e^{-\lambda^2 \xi^2}. \quad (4.59)$$

Integrating again,

$$F(\xi) = c_1 \int e^{-\lambda^2 \xi^2} d\xi + c_2. \quad (4.60)$$

where c_2 is another integration constant. Since the integral of $e^{-\lambda^2 \xi^2}$ is directly related

to the error function, we write:

$$\int e^{-\lambda^2 \xi^2} \partial \xi = \frac{\sqrt{\pi}}{2\lambda} \operatorname{erf}(\lambda \xi). \quad (4.61)$$

The error function is defined as:

$$\operatorname{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} dt. \quad (4.62)$$

Substituting,

$$F(\xi) = C_1 \frac{\sqrt{\pi}}{2\lambda} \operatorname{erf}(\lambda \xi) + C_2 \quad (4.63)$$

Applying boundary conditions: At $\xi = 0$, $F(0) = 1$,

$$F(0) = C_1 \frac{\sqrt{\pi}}{2\lambda} \operatorname{erf}(0) + C_2 = C_2 \quad (4.64)$$

Thus, $C_2 = 1$. At $\xi = 1$, $F(1) = 0$,

$$F(1) = C_1 \frac{\sqrt{\pi}}{2\lambda} \operatorname{erf}(\lambda) + C_2 \quad (4.65)$$

$$C_2 = -C_1 \frac{\sqrt{\pi}}{2\lambda} \operatorname{erf}(\lambda) \quad (4.66)$$

Substituting $C_2 = 1$,

$$1 = -C_1 \frac{\sqrt{\pi}}{2\lambda} \operatorname{erf}(\lambda) \quad (4.67)$$

$$C_1 = -\frac{2\lambda}{\sqrt{\pi} \operatorname{erf}(\lambda)} \quad (4.68)$$

Substituting C_1 and C_2 into the general solution,

$$F(\xi) = -\frac{2\lambda}{\sqrt{\pi} \operatorname{erf}(\lambda)} \frac{\sqrt{\pi}}{2\lambda} \operatorname{erf}(\lambda \xi) + 1 \quad (4.69)$$

Simplifying,

$$F(\xi) = \frac{1 - \operatorname{erf}(\lambda \xi)}{\operatorname{erf}(\lambda)} \quad (4.70)$$

This is the solution for the temperature profile in the liquid region.

4.2 Time Dependent Boundary Condition

This is a part of a mathematical analysis related to the Stefan problem, which typically models phase change phenomena, such as the melting of ice into water. specifically by this section discuss a time-dependent boundary condition for the problem.

The case (ii) in (2.20) $f(t) = e^t - 1$, is a time dependent condition and we need to make sure that $F(0, t)$ is independent of time in the limit $t \rightarrow 0$.

Boundary condition formula:

The boundary condition given is,

$$F = \frac{e^t - 1}{h(t)} \quad \text{at} \quad \zeta = 0. \quad (4.71)$$

To define $h(t)$, a Taylor expansion of $e^t - 1$ around $t = 0$ is performed. The exponential function e^t has a Taylor series expansion given by,

$$e^t = 1 + t + \frac{t^2}{2!} + \frac{t^3}{3!} + \frac{t^4}{4!} + \dots \quad (4.72)$$

When we subtract 1 from e^t , we get the Taylor series for $e^t - 1$:

$$e^t - 1 = [1 + t + \frac{t^2}{2!} + \frac{t^3}{3!} + \dots] - 1 = t + \frac{t^2}{2} + \frac{t^3}{6} + \dots$$

Thus the Taylor expansion of $e^t - 1$ around $t = 0$,

$$e^t - 1 = t + \frac{t^2}{2} + \frac{t^3}{6} + \dots$$

$$\text{ie, } f(t) \sim t + \frac{t^2}{2} + \frac{t^3}{6} + O(t^4). \quad (4.73)$$

The term $O(t^4)$ represents the higher order terms in the expansion that are of order t^4 or higher. These terms are neglected in this approximation because they contribute minimally as $t \rightarrow 0$.

The notation $O(t^4)$ is used to signify that the oriented terms are asymptotically

proportional to t^4 or smaller.

The boundary condition $h(t)$ was chosen to be $h(t) = t$, so that the leading order term t dominates as $t \rightarrow 0$. substituting $h(t) = t$ into the time dependent boundary condition (4.71), The Stefan condition becomes

$$s(t) \frac{ds}{dt} = O(t) \quad (4.74)$$

Now, Assume $O(t) \sim Ct$, where c is a constant as $t \rightarrow 0$. Substituting this into (4.74) we get:

$$s(t) \frac{ds}{dt} = Ct$$

Rewrite the entire equation as:

$$s(t)ds = Ctdt$$

Now integrate both sides:

Integrate with respect to t , assuming the initial condition $s(0) = 0$ (since the free boundary starts at zero at $t = 0$):

$$\int sds = \int Ctdt$$

This gives,

$$\frac{s^2}{2} = \frac{Ct^2}{2}$$

solve for $s(t)$, multiply through by 2 and take the square root:

$$s(t) = \sqrt{C}t$$

Introduce proportionality constant γ . Let $\gamma = \sqrt{C}$, which is a positive proportionality constant. The solution becomes:

$$s(t) = \gamma t \quad (4.75)$$

The system (4.20) ie,

$$h \frac{\partial^2 F}{\partial \zeta^2} = s \left[s \frac{dh}{dt} F + s h \frac{\partial F}{\partial t} - \zeta \frac{ds}{dt} - \frac{\partial F}{\partial \zeta} \right]$$

with $h(t)=1$ with in the limit $t \rightarrow 0$ be,

$$\frac{\partial^2 F}{\partial \zeta^2} = 0 \quad (4.76)$$

so, $\frac{\partial^2 F}{\partial \zeta^2} = 0$ [since $s(0)=0$]

subject to the boundary conditions are,

$$f(1) = 0, f(0) = 1, \quad \gamma^2 \beta = \frac{-\partial F}{\partial \zeta}, \quad \zeta = 1 \quad (4.77)$$

The differential equation is,

$$\frac{\partial^2 F}{\partial \zeta^2} = 0$$

This equation means that the second derivative of $F(\zeta)$ with respect to ζ is zero, indicating that $F(\zeta)$ is a linear function of ζ . Therefore, the general solution is,

$$F(\zeta) = C_1 \zeta + C_2$$

Where C_1 and C_2 are constants to be determined.

The boundary conditions are:

1. $F(0)=1$

$$2. F(1)=0$$

By Applying $F(0)=1$, Substitute $\zeta = 0$ into $F(\zeta) = C_1\zeta + C_2$

$$F(0) = C_1(0) + C_2 = 1 \rightarrow C_2 = 1$$

Applying $F(1) = 0$, substitute $\zeta = 1$ into $F(\zeta) = C_1\zeta + C_2$

$$F(1) = C_1(1) + C_2 = 0 \implies C_1 + 1 = 0 \implies C_1 = -1$$

substitute $C_1 = -1$ and $C_2 = 1$ into $F(\zeta) = C_1\zeta + C_2$:

$$F(\zeta) = -\zeta + 1$$

Rewriting this gives,

$$F(\zeta) = 1 - \zeta \tag{4.78}$$

To verify,

$$1. \text{ At } \zeta = 0 : F(0) = 1 - 0 = 1, \text{ which matches } F(0)=1$$

$$2. \text{ At } \zeta = 1 : F(1) = 1 - 1 = 0, \text{ which matches } F(1)=0$$

Thus the solution $F(\zeta) = 1 - \zeta$, satisfies the differential equation and the boundary conditions.

To solve equation (4.76) under these boundary conditions,

we proceed as follows:

Determining γ , The condition $\gamma^2\beta = \frac{-\partial F}{\partial \zeta}$ at $\zeta = 1$ requires us to find $\frac{\partial F}{\partial \zeta}$:

$$\frac{\partial F}{\partial \zeta} = -1$$

Substituting into the boundary condition at $\zeta = 1$,

$$\gamma^2\beta = -(-1) = 1$$

solving for γ :

$$\gamma \pm \frac{1}{\sqrt{\beta}}$$

According to the texts the positive solution is chosen, so

$$\gamma = \frac{1}{\sqrt{\beta}} \quad (4.79)$$

4.3 Numerical Methods for Solving Stefan Problems

As seen previously, the Crank-Nicholson method is a sufficient stable method for the heat equation. We will therefore use the Crank-Nicholson scheme to appropriate the solution of the Stefan problem under the assumption that the discretization of ξ and t is uniform.

4.3.1 Constant Boundary Condition

To solve the Stefan problem numerically, one common approach is to discrete the problem using finite differences which involves approximating the differential equations and boundary conditions at discrete points in both space and time.

The system can be re-written using $z = s^2(t)$, so that to simplify the free boundary representation. It will take the appearance:

$$\frac{\partial^2 F}{\partial \xi^2} = z \frac{\partial F}{\partial t} - \frac{\xi}{2} \frac{dz}{dt} \frac{\partial F}{\partial t}, \quad (4.80)$$

and the boundary and initial conditions will be:

$$\left\{ \begin{array}{ll} F = 0, & \text{at } \xi = 1 \\ \frac{\beta}{2} \frac{dz}{dt} = -\frac{\partial F}{\partial \xi}, & \text{at } \xi = 1 \\ F = 1, & \text{at } \xi = 0 \end{array} \right.$$

Now, we apply a discretization using the same method as in section(3.1.2) so that to acquire the Crank-Nicholson scheme. As we know, the Crank-Nicholson scheme approximates the time derivative by averaging the solution at the current and next time steps (implicit method), while also using central differences for the spatial derivatives: Discretization of variables is given by:

- Time t is divided into discrete steps: t^n, t^{n+1} .
- Space ξ is discretized into points ξ_j , where $j = 0, 1, 2, 3, \dots, M$.

The numerical approximations are:

$$F_j^n \approx F(\xi_j, t^n), \quad z^n \approx z(t^n).$$

Then by applying the Crank-Nicholson method as in section(3.1.2), we get:

$$\begin{aligned} & \frac{1}{2} \left(\frac{F_{j+1}^{n+1} - 2F_j^{n+1} + F_{j-1}^{n+1}}{h^2} \right) + \left(\frac{F_{j+1}^n - 2F_j^n + F_{j-1}^n}{h^2} \right) \\ &= z^{n+\frac{1}{2}} \left(\frac{F_j^{n+1} - F_j^n}{k} \right) - \frac{\xi_j}{2} \frac{z^{n+1} - z^n}{k} \left[\frac{1}{2} \left(\frac{F_{j+1}^{n+1} - F_{j-1}^{n+1}}{2h} \right) + \frac{1}{2} \left(\frac{F_{j+1}^n - F_{j-1}^n}{2h} \right) \right], \end{aligned} \quad (4.81)$$

where,

$$\begin{cases} h & : \text{spatial step size} \\ k & : \text{time step size} \\ z^{n+1} & = \frac{z^{n+1} + z^n}{2} \end{cases}$$

The scheme uses a central difference at time $t^{n+\frac{1}{2}}$ and a second order central difference for the spatial derivative at position ξ_j .

So, the boundary conditions can be discretized as follows:

$$\begin{cases} F_0^{n+1} = 1, & \text{at } \xi = 0 \ (j = 0) \\ F_m^{n+1} = 0, & \text{at } \xi = 1 \ (j = 1) \\ \frac{\beta}{2} \left(\frac{z^{n+1} - z^n}{k} \right) = -\frac{1}{2} \left(\frac{F_{m+1}^{n+1} - F_{m-1}^{n+1}}{2h} \right) - \frac{1}{2} \left(\frac{F_{M+1}^n - F_{M-1}^n}{2h} \right) \end{cases} \quad (4.82)$$

Moving forward, to be able to approximate this numerically, we need to write the discretized formula in matrix form. We know that the Crank-Nicholson scheme leads to a matrix equation for the unknown F^{n+1} at the next time step:

$$LF^{n+1} = RF^n - 2C,$$

where,

- L: coefficient matrix for the left-hand side (terms at $n + 1$).
- R: coefficient matrix for the right-hand side (terms at n).
- C: a vector accounting for boundary conditions.

We can re-arrange the discretized formula as follows:

$$\tilde{L} = \frac{1}{2h^2}(F_{j+1}^{n+1} - 2F_j^{n+1} + F_{j-1}^{n+1}) - \frac{z^{n+\frac{1}{2}}}{k}F_j^{n+1} + \frac{z'}{8h}\xi_j(F_{j+1}^{n+1} - F_{j-1}^{n+1}), \quad (4.83)$$

and

$$\tilde{R} = -\frac{1}{2h^2}(F_{j+1}^n - 2F_j^n + F_{j-1}^n) - \frac{z^{n+\frac{1}{2}}}{k}F_j^n \xi_j(F_{j+1}^n - F_{j-1}^n), \quad (4.84)$$

where,

$$z' = \frac{z^{n+1} - z^n}{k} \quad z^{n+\frac{1}{2}} = \frac{z^{n+2} + z^n}{2}. \quad (4.85)$$

To take care if the boundary condition $F_0^{n+1} = 1$, we evaluate equation 4.83 (equation 2 is also done in the same way) at $j = 1$:

$$\tilde{L}_{j=1} = \frac{1}{2h^2}(F_2^{n+1} - 2F_1^{n+1} + F_0^{n+1}) - \frac{z^{n+\frac{1}{2}}}{k}F_1^{n+1} + \frac{z'}{8k}\xi_1(F_2^{n+1} - F_0^{n+1}). \quad (4.86)$$

Sorting out the two constant vectors (due to the boundary condition) we could put

them together into one constant vector. ie,

$$C \equiv \begin{pmatrix} \frac{1}{2h^2} - \frac{z'}{8h}\xi_1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}$$

Now the coefficient matrix of the first paranthesis in equation 4.83 is given by:

$$A \equiv \frac{1}{2h^2} \begin{pmatrix} -2 & 1 & 0 & \dots & 0 \\ 1 & -2 & 1 & \ddots & \vdots \\ 0 & \ddots & \ddots & \ddots & 0 \\ \vdots & \ddots & 1 & -2 & 1 \\ 0 & \dots & 0 & 1 & -2 \end{pmatrix}$$

and the second coefficient matrix equation is:

$$B = \frac{1}{8h} \begin{pmatrix} 0 & \xi_1 & 0 & \dots & \dots & \dots & 0 \\ -\xi_2 & 0 & \xi_2 & \ddots & \dots & \dots & \vdots \\ 0 & -\xi_3 & 0 & \xi_3 & \dots & \dots & 0 \\ 0 & \ddots & \ddots & \ddots & \ddots & \ddots & 0 \\ \vdots & \ddots & \ddots & \ddots & -\xi_{M-2} & 0 & \xi_{M-2} \\ 0 & \dots & \dots & \dots & 0 & -\xi_{M-1} & 0 \end{pmatrix}$$

Then the system can be written in matrix form as:

$$\left(A - \frac{2^{n+\frac{1}{2}}}{k}I + z'B\right)F^{n+1} = \left(-\frac{z^{n+\frac{1}{2}}}{k}I - z'B - A\right)F^n - 2C, \quad (4.87)$$

where I is the $(M-1)$ identity matrix. ie., System can be formulated in matrix form as:

$$LF^{n+1} = RF^n - 2C \quad (4.88)$$

where,

$$L = (A - \frac{z^{n+\frac{1}{2}}}{k}I + z'B), \quad R = (-\frac{z^{n+\frac{1}{2}}}{k}I - z'B - A) \quad (4.89)$$

and

$$C \equiv \begin{pmatrix} \frac{1}{2h^2} - \frac{z'}{8h}\xi_1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}$$

The approximated Stefan condition in equation 4.82 include the extrapolated value F_{M+1}^{n+1} and F_{M+1}^n . To approximate this problem we evaluate equation 4.81 at $j = M$ and thus we can eliminate the extrapolated value by some algebra. This will lead us to the following polynomial equation of second order for z^{n+1}

$$\begin{aligned} & \frac{\beta\xi_M}{2k}(z^{n+1})^2 + [-\frac{\beta\xi_M}{k}z^n + \frac{2\beta r}{v} + F_M^{n+1} - F_M^n]z^{n+1} \\ & + \frac{\beta\xi_M}{2k}(z^n)^2 - \frac{2\beta r}{v}z^n + (2r + z^n)F_M^n - 2r(F_{M-1}^{n+1} + F_{M-1}^n) = 0. \end{aligned} \quad (4.90)$$

Algorithm *The constant boundary condition*

1. Start by introducing all the constants, "Allocate" memory for the vectors, and give the initial conditions their values.
2. From equation 2.80 we can use the Newton-Raphson method to approximate the physical constant λ related to the given value of β .
3. Start iterating a time loop for the whole problem. Because equation 4.81 involves z at time level $(n + 1)$, we need to iterate the value until a given tolerance is reached.
4. We use the starting guess for the free boundary to be the previous value $z(n)$ and with help of 4.90 we can update (z^{n+1}) until the tolerance is reached.
5. Convert back the values for the temperature and the free boundary. Save values at each time step for the exact solutions for both the free boundary and the temperature, for later usage.
6. Plot both the numerical and the exact values for both the temperature and the free boundary against time. Plot the error in the numerical temperature and the numerical free boundary against time.

4.3.2 Time Dependent Boundary Condition

For the second case in equation (4.21), $h(t) = t$

$$\left\{ \begin{array}{ll} t \frac{\partial^2 F}{\partial \xi^2} = s[sF + st \frac{\partial F}{\partial t} - \xi t \frac{\partial s}{\partial t} \frac{\partial F}{\partial \xi}], & at\xi = 1 \\ F = 0 & at\xi = 1 \\ \beta s \frac{\partial s}{\partial t} = -t \frac{\partial F}{\partial \xi} & at\xi = 1 \\ F = \frac{e^t - 1}{t} & \xi = 0 \end{array} \right. \quad (4.91)$$

The time-dependent boundary condition for the second case in the equation is considered, where $h(t)$ is given by a set of boundary conditions. The governing equation is presented as,

$$\frac{\partial^2 F}{\partial \xi^2} = s \left[F + st \frac{\partial F}{\partial t} - \xi ts \frac{ds}{dt} \frac{\partial F}{\partial \xi} \right], \quad (4.92)$$

with three key terms on the right-hand side,

- Source Term : $s^2 F$
- Time Derivative Term : $s^2 t \frac{\partial F}{\partial t}$
- Convective Term : $-\xi ts \frac{ds}{dt} \frac{\partial F}{\partial \xi}$

The Crank-Nicolson discretization method is applied, which involves averaging between time steps n and $n + 1$, leading to the approximations,

$$\frac{\partial F}{\partial t} \approx \frac{F^{n+1} - F^n}{\Delta t}, \quad (4.93)$$

$$\frac{\partial^2 F}{\partial \xi^2} \approx \frac{F_{j+1}^{n+1} - 2F_j^{n+1} + F_{j-1}^{n+1}}{h^2}, \quad (4.94)$$

$$\frac{\partial F}{\partial \xi} \approx \frac{F_{j+1}^{n+1} - F_{j-1}^{n+1}}{2h}. \quad (4.95)$$

Subsequently, each term is discretized separately. The spatial derivative term (left-hand side) is rewritten as,

$$t^{n+\frac{1}{2}} \frac{\partial^2 F}{\partial \xi^2}, \quad (4.96)$$

which is further discretized using Crank-Nicolson averaging. The source term is discretized as,

$$s^2 F = \left(s^{n+\frac{1}{2}} \right)^2 \frac{1}{2} (F_j^{n+1} + F_j^n), \quad (4.97)$$

the time derivative term as,

$$s^2 t \frac{\partial F}{\partial t} = s^2 t \frac{F_j^{n+1} - F_j^n}{\Delta t}, \quad (4.98)$$

and the convective term as,

$$-\xi t s \frac{ds}{dt} \frac{\partial F}{\partial \xi} \approx -\xi_j t^{n+\frac{1}{2}} s^{n+\frac{1}{2}} \frac{ds}{dt} \frac{1}{2h} (F_{j+1}^{n+1} - F_{j-1}^{n+1} + F_{j+1}^n - F_{j-1}^n). \quad (4.99)$$

Combining the discretized terms results in,

$$\begin{aligned} & \frac{1}{2}(t_{n+1} + t_n) \left[\frac{F_{j+1}^{n+1} - 2F_j^{n+1} + F_{j-1}^{n+1}}{h^2} + \frac{F_{j+1}^n - 2F_j^n + F_{j-1}^n}{h^2} \right] = \\ & (s^{n+1/2}) \frac{1}{2} (F_j^{n+1} + F_j^n) + (s^{n+1/2}) \frac{F_j^{n+1} - F_j^n}{\Delta t} - \xi_j^{n+1/2} s^{n+1/2} \frac{1}{2} \left[\frac{F_{j+1}^{n+1} + F_{j+1}^n}{2h} + \frac{F_{j-1}^{n+1} - F_{j-1}^n}{2h} \right] \end{aligned} \quad (4.100)$$

Now, we need to apply the Crank-Nicholson method to the Stefan problem with time-dependent boundary conditions at $\xi = 0$ and $\xi = 1$. At $\xi = 0$, the boundary condition in the transformed co-ordinate is given by,

$$F(0, t) = \frac{e^t - 1}{t}.$$

At the discretized grid point $\xi = 0$, this becomes,

$$F_0^n = \frac{e^{t^n} - 1}{t^n}, \quad F_0^{n+1} = \frac{e^{t^{n+1}} - 1}{t^{n+1}},$$

where F_0^n is the value of F at the current time step t_n , and F_0^{n+1} is the next time step t_{n+1} .

Using the Crank-Nicholson approximation,

$$F_0^{n+\frac{1}{2}} = \frac{1}{2} (F_0^n + F_0^{n+1}).$$

Substituting the values for F_0^{n+1} and F_0^n using the boundary condition expression,

$$F_0^{n+\frac{1}{2}} = \frac{1}{2} \left(\frac{e^{t^{n+1}} - 1}{t_{n+1}} + \frac{e^{t^n} - 1}{t_n} \right).$$

Thus the equation becomes,

$$\frac{1}{2} (F_0^n + F_0^{n+1}) = \frac{1}{2} \left(\frac{e^{t^{n+1}} - 1}{t_{n+1}} + \frac{e^{t^n} - 1}{t_n} \right), \quad (4.101)$$

$s(\xi, t)$ at $\xi = 1$,

$$\beta \frac{dS}{dt} = -t \frac{\partial F}{\partial \xi} \Big|_{\xi=1}. \quad (4.102)$$

Discretizing $\frac{dS}{dt}$, using a backward difference for the time derivative,

$$\frac{dS}{dt} \approx \frac{S^{n+1} - S^n}{k},$$

where k is the time step size.

Discretizing $\frac{\partial F}{\partial \xi}$. Using a central difference for $\frac{\partial F}{\partial \xi}$ at $\xi = 1$,

$$\frac{\partial F}{\partial \xi} \Big|_{\xi=1} \approx \frac{1}{2} \left(\frac{F_{M+1}^{n+1} - F_{M-1}^{n+1}}{2h} + \frac{F_{M+1}^n - F_{M-1}^n}{2h} \right), \quad (4.103)$$

where h is the spatial step size. Substituting this into the Stefan condition:

$$\beta S^{n+1} \frac{S^{n+1} - S^n}{k} = -t^{n+\frac{1}{2}} \frac{1}{2} \left(\frac{F_{M+1}^{n+1} - F_{M-1}^{n+1}}{2h} + \frac{F_{M+1}^n - F_{M-1}^n}{2h} \right). \quad (4.104)$$

Dirichlet Condition at the Boundary $\xi = 1$,

$$F(1, t) = 0. \quad (4.105)$$

Discretizing the Condition At the spatial grid point $M(\xi = 1)$,

$$F_M^{n+1} = 0. \quad (4.106)$$

The boundary conditions become,

$$\left\{ \begin{array}{l} \beta S^{n+\frac{1}{2}} s' = -\frac{1}{2} t^{n+\frac{1}{2}} \left(\frac{F_{M+1}^{n+1} - F_{M-1}^{n+1}}{2h} \right) - \frac{1}{2} t^{n+\frac{1}{2}} \left(\frac{F_{M+1}^n - F_{M-1}^n}{2h} \right), \\ \frac{1}{2} (F_0^{n+1} + F_0^n) = \frac{1}{2} \left(\frac{e^{t^{n+1}} - 1}{t^{n+1}} + \frac{e^{t^n} - 1}{t^n} \right), \\ F_M^{n+1} = 0. \end{array} \right. \quad (4.104)$$

The system (4.100) involves s at the next time level, so an iteration is needed. We use the value at level n as a starting guess for s^{n+1} . To update the value of s^{n+1} , we use the Stefan condition in (4.104). However, equation (4.104) involves the two extrapolated values F_{M+1}^n and F_{M+1}^{n+1} . To remove those values, we evaluate equation (4.100) at $j = M$, leading to a quartic expression in terms of s^{n+1} . After discretizing the second derivative we have,

$$\frac{\partial^2 F}{\partial \xi^2} \Big|_M = \frac{F_{j+1}^n - 2F_j^n + F_{j-1}^n}{h^2}, \quad (4.107)$$

using Crank-Nicholson method for $\frac{\partial^2 F}{\partial \xi^2}$,

$$t^{n+\frac{1}{2}} \left[\frac{1}{2} \frac{F_{m+1}^{n+1} - 2F_m^{n+1} + F_{m-1}^{n+1}}{h^2} + \frac{1}{2} \frac{F_{m+1}^n - 2F_m^n + F_{m-1}^n}{h^2} \right].$$

Removing F_{M+1}^{n+1} and F_{M-1}^{n+1} . Combine the Stefan condition equation (4.104) to eliminate these terms. This results in a quartic equation for s^{n+1} . We are assuming $F_M^{n+1} = 0$ (because of the boundary condition). This introduces terms involving s^{n+1} and s^n . After substituting the discretizations into the equation and removing

extrapolated terms:

1. Combine terms proportional to $(s^{n+1})^2$ and $(s^n)^2$.
2. Include contributions from the Stefan condition to replace derivatives of s with known finite differences.

The equation becomes,

$$\left[2r(t^{n+1} + t^n)(F_{M-1}^{n+1} + F_{M-1}^n) - 4\beta \left[r + \frac{\xi_M}{4h} \right] [(s^{n+1})^2 - (s^n)^2] \right] [(s^{n+1})^2 - (s^n)^2] = 0. \quad (4.108)$$

- r and β are constants.
- F_{M-1}^{n+1} and F_{M-1}^n are the values of F at the last known grid point.
- ξ_M and h are spatial parameters.

To be able to approximate this numerically we need to write (4.100) in matrix form, and take into account that,

$$\frac{1}{2} (F_0^{n+1} + F_0^n) = \frac{1}{2} \left(\frac{e^{t^{n+1}} - 1}{t^{n+1}} + \frac{e^{t^n} - 1}{t^n} \right),$$

at $\xi = 0$. Denote the left-hand side of equation (4.100) as LH and evaluate at $j = 1$.

$$LH_{j=1} = \frac{t^{n+\frac{1}{2}}}{2h^2} [F_2^{n+1} - 2F_1^{n+1} + F_0^{n+1} + F_2^n - F_1^n + F_0^n], \quad (4.109)$$

and the right-hand side of (4.100) at $j = 1$,

$$RH_{j=1} = \left(s^{n+\frac{1}{2}} \right)^2 F_1^{n+\frac{1}{2}} + t^{n+\frac{1}{2}} \left(s^{n+\frac{1}{2}} \right)^2 \frac{F_1^{n+1} - F_1^n}{k} - \frac{\xi_1 t^{n+\frac{1}{2}} s^{n+\frac{1}{2}} s'}{4h} [F_2^{n+1} - F_1^{n+1} + F_2^n - F_1^n]. \quad (4.110)$$

The boundary conditions contribute constant terms to the system. This is moved to a vector $D(t)$. $D(t)$ is the contributions from F_0^{n+1} and F_0^n are grouped into a vector $D(t)$, which incorporates the boundary condition at $\xi = 0$. This vector accounts for the impact of the boundary on all grid points. Specifically, at $j = 1$, the term $\frac{e^t - 1}{t}$ introduces a constant vector contribution. For example,

$$D_1(t) = \left(\frac{t^{n+\frac{1}{2}}}{2h^2} - \frac{\xi_1 t^{n+\frac{1}{2}} s^{n+\frac{1}{2}} s'}{4h} \right) \left(\frac{e^{t^{n+1}} - 1}{t^{n+1}} + \frac{e^{t^n} - 1}{t^n} \right).$$

Similarly, for other nodes, $D(t)$ will have zero entries except where the boundary condition indirectly influences the system. After substituting the boundary conditions and simplifying, the system takes the form,

$$LF^{n+1} = RF^n + D(t),$$

where,

- L is the tridiagonal matrix for the current time step coefficients.
- R is the tridiagonal matrix for the previous time step coefficients.
- $D(t)$ is a vector containing contributions from the boundary condition.

And with,

$$L = t^{n+\frac{1}{2}}A - (s^{n+\frac{1}{2}})^2 \left(\frac{1}{2}x + \frac{1}{t^{n+1}} \right) I + t^{n+\frac{1}{2}}s^{n+\frac{1}{2}}s'B,$$

$$R = -t^{n+\frac{1}{2}}A + (s^{n+\frac{1}{2}})^2 \left(\frac{1}{2} - \frac{1}{t^n} \right) I - t^{n+\frac{1}{2}}s^{n+\frac{1}{2}}s'B,$$

where I is the $(M - 1) \times (M - 1)$ identity matrix, and the other coefficients are:

$$A = \frac{1}{2h^2} \begin{bmatrix} -2 & 1 & 0 & \cdots & 0 \\ 1 & -2 & 1 & \cdots & 0 \\ 0 & 1 & -2 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 1 & \cdots & -2 \end{bmatrix},$$

$$B = \frac{1}{4h} \begin{bmatrix} 0 & \xi_1 & 0 & \cdots & 0 \\ -\xi_2 & 0 & \xi_2 & \cdots & \\ 0 & -\xi_3 & 0 & \xi_3 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ \vdots & \vdots & -\xi_{m-2} & 0 & -\xi_{m-2} \\ \vdots & \vdots & 0 & -\xi_{m-2} & \ddots 0 \end{bmatrix},$$

$$D(t) = \begin{bmatrix} \left(\frac{t^{n+\frac{1}{2}}}{2h^2} - \frac{\xi_1 t^{n+\frac{1}{2}} s^{n+\frac{1}{2}} s'}{4h} \right) \left(\frac{e^{t^{n+1}} - 1}{t^{n+1}} + \frac{e^{t^n} - 1}{t^n} \right) \\ 0 \\ \vdots \\ 0 \end{bmatrix}.$$

Algorithm *The time-dependent boundary condition*

1. Start by introducing all the constants, “Allocate ”memory for the vectors, and give the initial conditions their values.
2. Start iterating a time loop for the whole problem, and because equation (4.100) involves s at time level $(n + 1)$, we need to iterate the value until a given tolerance is reached for every time step.
3. Take care of the case with $0/0$ separately at $n = 1$, where the values in the limit $t \rightarrow 0$ is analyzed previously in section (??).
4. Due to boundary condition at $\xi = 0$ we get a time constant vector D for every time step.
5. Introduce the right and left coefficient matrix for the expression $LF^{n+1} = RF^n + D$ and solve for F^{n+1} .
6. Solve the quartic equation with Newton-Raphson method with initial guess of $s^{n+1} = s^n$.
7. When a sufficient good value of s^{n+1} is found we add the boundary conditions at $\xi = 0$ and $\xi = 1$. Avoid the $0/0$ case at $n = 1$.
8. Convert back the values for the temperature and the free boundary. Save values at each time step for the exact solutions for both the free boundary and the temperature, for later usage.
9. Plot both the numerical and the exact values for both the temperature and the free boundary against time. Plot the error in the numerical temperature and the numerical free boundary against time.

4.4 Error Analysis

In this section we would like to discuss the methods used around the numerical analysis to measure the error. To evaluate the error in the numerical solution for temperature, the discrete l_2 norm $\|\cdot\|_{2,n}$ is used. At a time t_n , the l_2 norm error E_n is calculated as,

$$E_n = \|U^n - u^n\|_{2,h} = (h \sum_{j=0}^M (U(x_j, t_n) - U_j^n)^2)^{1/2}, \quad (4.111)$$

where, $U(x_j, t_n)$ is the exact solution at spatial point x_j and time t_n . U_j^n is the numerical solution at the same point. The error in the position of the free boundary S at time t_n is given by,

$$|S(t_n) - S_n|. \quad (4.112)$$

Where, $S(t_n)$ represents the exact position of the free boundary at t_n , S_n is the numerical approximation for the boundary position. System(4.81) involves Z^{n+1} and system (4.100) involves S^{n+1} . To be able to get those value we do an iteration on Z^{n+1} respectively S^{n+1} for a given tolerance ϵ using the same value at level n as a starting guess. We get the new update from the Stefan condition (4.82), and keep on updating until the tolerance is reached. If we denote Z_m^{n+1} and S_m^{n+1} as the $m : t_n$ iterated value of Z^{n+1} and S^{n+1} , we can write a convergence criterion as,

$$|Z_{m+1}^{n+1} - Z_m^{n+1}| < \epsilon \quad |S_{m+1}^{n+1} - S_m^{n+1}| < \epsilon. \quad (4.113)$$

Chapter 5

Conclusion

In this thesis we have introduced the one-dimensional Stefan problem defined on a semi-infinite interval with two different type of boundary conditions. The important Stefan condition for the free boundary was derived under the condition of equal densities for the two phases. That is not realistic, and even for water the small difference will give a volume change in the transition. Due to the change in volume, another term will occur in the heat equation and also change the appearance for the Stefan condition. By taking consideration of this small but realistic feature, the whole problem of ours will get more complex.

Another simplification in this project was the start temperature of the solid phase. If the (for our case) solid phase had another temperature then the melting point, a heat flux will flow back into the liquid phase. And thus attain another term for the Stefan condition (seen in equation 2.1) further more we will need to solve two different heat equation, separately describing the temperature in each phase.

The numerical simulation was made with the Crank-Nicholson method, which have a second order accuracy in both time and space. This method uses an implicit discretization about $(n + \frac{1}{2})$, instead of a more straight forward explicit one. The numerical implementation is harder, but as shown previously. The explicit scheme imposes both lesser accuracy and a restriction of our time step.

5.1 Future Studies

It does clearly exist a lot of opportunities to continue with similar problems That i have been working on this semester. In this project only one time-dependent boundary condition was analyzed and it does exist some other more realistic and existing condition, for instance;oscillating function which could be describing the variation of temperature over a day. Evan further if could be possible to expand the model. Find an adequate real life problem. It would be vary interesting to simulate nature itself.

Furthermore, this was a one-phase problem so we could extend our interest for a problem with the temperature changing over time in both of the phase, two-phase problem. So the temperature would change in both the liquid phase and the solid phase. For any problem we have discussed in this project, we have skipped the density difference between the phases, which causes the liquid phase to attain another velocity vector due to volume change in the phase transition.

For the confident and brave person. The option to go up a dimension does always exist .Look at some weak solution and maybe discover something exciting. But as far as that I have understood. The level of complexity do increases substantially as we go up in dimensions.

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